DISTRICT PANCHAYATH KASARAGOD

EQIP 2026

Educational Quality Improvement Programme for class -X

BIOLOGY

ENGLISH MEDIUM

Standard -X

QUESTION POOL



DISTRICT INSTITUTE OF EDUCATION AND TRAINING (DIET)
KASARAGOD

EQIP 2025-26

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ആശംസ

വികേന്ദ്രീകൃത ആസൂത്രണത്തിലൂടെയും നിർവ്വഹണത്തിലൂടെയും കേരളത്തിലെ ആരോഗ്യവിദ്യാഭ്യാസ മേഖലകളെ ദേശീയ തലത്തിൽ ഒന്നാമതെത്തിക്കാൻ നമുക്ക് കഴിഞ്ഞിട്ടുണ്ട്. ഈ നേട്ടങ്ങൾ കൈവരിക്കാൻ പ്രാദേശിക ഭരണകൂടങ്ങൾ സ്തുത്യർ ഹമായ പങ്കുവഹിച്ചു. ദേശീയ സംസ്ഥാനതല പഠനങ്ങൾ നമ്മുടെ കുട്ടികളുടെ പഠന നിലവാരം ഇനിയും ഉയരേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറെ വർഷങ്ങളായി കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ പൊതുവിദ്യാഭ്യാസ വകുപ്പും കാസറഗോഡ് ഡയറ്റും ഒത്തു ചേർന്ന് നടപ്പാക്കുന്ന **'എക്വിപ്പ്' 2025-26 (EQIP 2025-26)** പഠനപരിപോഷണ പരിപാടിക്ക് എല്ലാ പിന്തുണയും ഉറപ്പുതരുന്നു. പൊതുപരീക്ഷകളെ അഭിമുഖീകരിക്കുന്ന പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രവർത്തനങ്ങളുടെ മികവിന്റെ അടയാളമായി മാറുകയാണ് വാർഷിക പരീക്ഷകൾ. അറിവിന്റെ തെളിമയോടെ ഓരോ വിദ്യാർത്ഥിക്കും പരീക്ഷ എഴുതാൻ കഴിയണം. വിദ്യാർത്ഥികളുടെ ജീവിതത്തിലെ ഏറ്റവും പ്രധാനപ്പെട്ട പരീക്ഷകൾക്ക് വേണ്ടി തയ്യാറാക്കിയ ചോദ്യബാങ്കിന് എല്ലാവിധ ആശംസകളും നേരുന്നു. നന്നായി പഠിക്കുക. പരീക്ഷയെ സധൈര്യം നേരിടുക. തളരാതെ മുന്നോട്ട്. വിജയം നിങ്ങളോടൊപ്പമുണ്ട്. ആശംസകൾ.

ശ്രീമതി ബേബി ബാലകൃഷ്ണൻ
ജില്ലാ പഞ്ചായത്ത് പ്രസിഡന്റ്
കാസറഗോഡ്



ആശംസ

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസ മേഖലയെ ശക്തിപ്പെടുത്തുന്നതിന് നിരവധി പ്രവർത്തനങ്ങളാണ് നടന്നുവരുന്നത്. പൊതുവിദ്യാഭ്യാസ മേഖലയെ പൂർവ്വാധികം കരുത്തോടെ നാം മുന്നോട്ട് നയിക്കുകയാണ്. ഈ ഘട്ടത്തിലാണ് കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തും, പൊതുവിദ്യാഭ്യാസ വകുപ്പും, വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രവും **(DIET)** പത്താം ക്ലാസ് വിദ്യാർത്ഥികളുടെ പഠന വിടവുകൾ പരിഹരിക്കുന്നതിനും ആത്മവിശ്വാസത്തോടെ പൊതുപരീക്ഷയെ നേരിടാൻ അവരെ പ്രാപ്തരാക്കുന്നതിനും വേണ്ടി പഠനപരിപോഷണ സാമഗ്രി തയ്യാറാക്കിയിട്ടുള്ളത്. ഈ പദ്ധതിയുടെ ഭാഗമായി ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ചുള്ള ചോദ്യശേഖരമാണ് **(Question Pool)** ഈ അക്കാദമിക വർഷം തയ്യാറാക്കിയിട്ടുള്ളത്. കുട്ടികളുടെ അക്കാദമിക് മികവ് ഉറപ്പുവരുത്തിക്കൊണ്ട് മികച്ച ഗ്രേഡുകൾ നേടാൻ അവരെ സജ്ജമാക്കാൻ **എക്വിപ് 2025-26 (EQIP-2025-26)** എന്ന പേരിൽ തയ്യാറാക്കിയ ഈ പദ്ധതിക്ക് കഴിയട്ടെയെന്ന് ആശംസിക്കുന്നു.

സ്നേഹപൂർവ്വം

അഡ്വ. സരിത എസ്.എൻ.

ആരോഗ്യ-വിദ്യാഭ്യാസ സ്ഥിരം സമിതി അധ്യക്ഷ,

ജില്ലാ പഞ്ചായത്ത്, കാസറഗോഡ്



ആശംസ

ജില്ലയിലെ അക്കാദമിക പ്രവർത്തനങ്ങളെ ഏകോപിപ്പിച്ച് മുന്നോട്ട് നയിക്കുന്നതിന്റെ പ്രധാന ഉത്തരവാദിത്തം കാസറഗോഡ് ജില്ലാ വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രത്തിന്റെ (DIET) നേതൃത്വത്തിൽ കാലങ്ങളായി ചെയ്തുവരുന്നുണ്ട്. പ്രീ-പ്രൈമറി തലം മുതൽ ഹയർസെക്കണ്ടറി തലംവരെയുള്ള മേഖലകളിൽ വിവിധങ്ങളായ പദ്ധതികൾ ആസൂത്രണം ചെയ്യാനും നിർവഹിക്കാനും ഡയറ്റിന് ഇതിനകം കഴിഞ്ഞിട്ടുണ്ട്. പത്താം ക്ലാസ്സിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറച്ചു വർഷങ്ങളായി ഡയറ്റിന്റെ നേതൃത്വത്തിൽ വിദ്യാഭ്യാസ വകുപ്പ് നടപ്പിലാക്കുന്ന പദ്ധതിയാണ് **എക്വിപ് (Educational Quality Improvement Programme).** അതാതു വർഷത്തെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഗണിച്ചുകൊണ്ടാണ് പ്രവർത്തനങ്ങൾ ആസൂത്രണം ചെയ്യുന്നത്. പത്താംതരം പാഠപുസ്തകവും വിലയിരുത്തൽ രീതിയും മാറിയ ഘട്ടത്തിൽ ആണല്ലോ നമ്മൾ. അതു കൊണ്ട് ഈ പദ്ധതിയുടെ ഭാഗമായി പത്താംതരത്തിലെ ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ പുതിയ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ച് ചോദ്യശേഖരം (Question Pool) പരിചയപ്പെടുത്താനാണ് ഇപ്രാവശ്യം ഉദ്ദേശിക്കുന്നത്. ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ഡയറ്റിന്റെ അക്കാദമിക സഹായത്തോടെ മലയാളത്തിലും ഇംഗ്ലീഷിലും കന്നഡയിലും പത്താം തരത്തിലെ ഐ.ടി. ഒഴികെയുള്ള വിഷയങ്ങളിൽ ചോദ്യശേഖരം തയ്യാറാക്കി നൽകാനാണ് ഉദ്ദേശിക്കുന്നത്. ഈ അധ്യയനവർഷം പത്താംതരം പരീക്ഷ എഴുതുന്ന മുഴുവൻ കുട്ടികൾക്കും ഈ പദ്ധതിയുടെ പ്രയോജനം ലഭിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു. വ്യത്യസ്ത പഠനശേഷിയും പഠനമികവുമുള്ള എല്ലാ വിഭാഗം കുട്ടികൾക്കും ഈ സാമഗ്രി പ്രയോജനപ്പെടട്ടെ എന്ന് ആശംസിക്കുന്നു. ഒരു കൂട്ടം അധ്യാപകരുടെ ആത്മാർത്ഥമായ പ്രവർത്തനമാണ് ഈ സാമഗ്രി വികസിപ്പിക്കാൻ സഹായകമായത്. അധ്യാപകരുടെ ആത്മാർത്ഥമായ പിന്തുണയും പ്രോത്സാഹനവും അനിവാര്യമായ ഈ ഉദ്യമത്തിൽ എല്ലാവരുടെയും സഹായസഹകരണങ്ങൾ തുടർന്നും പ്രതീക്ഷിച്ചുകൊണ്ട് ഏവർക്കും വിജയാശംസകൾ നേരുന്നു. ആശംസകളോടെ,

ഡോ. രഘുരാമ ഭട്ട് കെ.
പ്രിൻസിപ്പാൾ
ഡയറ്റ് കാസറഗോഡ്



ആമുഖം

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസം ശക്തിപ്പെടുത്തുന്നതിന് വ്യത്യസ്തങ്ങളായ നിരവധി പ്രവർത്തനങ്ങൾ നടന്നുവരികയാണ്. അതേസമയം ദേശീയ-സംസ്ഥാന പഠനങ്ങൾ നമ്മുടെ ജില്ലയിലെ കുട്ടികളുടെ പ്രകടനം ഇനിയും മെച്ചപ്പെടേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. ഈ പശ്ചാത്തലത്തിലാണ് ജില്ലയിലെ പൊതുവിദ്യാലയങ്ങളിൽ നിന്ന് **2025-26** അധ്യയന വർഷം എസ്.എസ്.എൽ.സി. പരീക്ഷകൾ അഭിമുഖീകരിക്കുന്ന കുട്ടികൾക്ക് പഠനപിന്തുണ നൽകുന്നതിന് ജില്ലാ പഞ്ചായത്തിന്റെയും പൊതുവിദ്യാഭ്യാസ വകുപ്പിന്റെയും സംയുക്താഭിമുഖ്യത്തിൽ വ്യത്യസ്ത വിഷയങ്ങളിൽ പഠനസാമഗ്രികൾ തയ്യാറാക്കേണ്ടതിന്റെ ആവശ്യകത ജില്ലാതല ഉന്നതാധികാര യോഗങ്ങളിൽ ചർച്ചചെയ്യപ്പെട്ടത്. ഇതിന്റെ അടിസ്ഥാനത്തിൽ ഡയറ്റ് കാസറഗോഡിന്റെ അക്കാദമിക നേതൃത്വത്തിൽ ജില്ലയിലെ മികച്ച അധ്യാപകരെ ഉൾപ്പെടുത്തിക്കൊണ്ട് പത്താംതരത്തിൽ ഐ.ടി. ഒഴിച്ചുള്ള എല്ലാ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിലനുസരിച്ച് ചോദ്യശേഖരം **(Question Pool)** തയ്യാറാക്കിയിരിക്കുകയാണ്. ജില്ലയിൽ നിന്നും പൊതുപരീക്ഷയെ അഭിമുഖീകരിക്കുന്ന മുഴുവൻ എസ്.എസ്.എൽ.സി, വിദ്യാർത്ഥികൾക്കും ആത്മവിശ്വാസം വളർത്തുന്നതിനും ഉന്നത വിജയം നേടുന്നതിനും ഈ ഉദ്യമം സഹായകമാകട്ടെയെന്ന് ആത്മാർത്ഥമായി ആഗ്രഹിക്കുന്നു. ഈ പദ്ധതിയെ നെഞ്ചേറ്റിയ പ്രിയപ്പെട്ട അധ്യാപക സുഹൃത്തുക്കൾക്ക് ഈ പുസ്തകത്തെ ഫലപ്രദമായി ഉപയോഗിക്കാൻ കഴിയട്ടെ.

എല്ലാവർക്കും വിജയാശംസകൾ.

ശ്രീ. മധുസൂദനൻ ടി. വി.
ജില്ലാ വിദ്യാഭ്യാസ ഉപഡയറക്ടർ
കാസറഗോഡ്

BIOLOGY

English Medium

Chapter 1

GENETICS OF LIFE

1 Mark Questions

1. Read the Statement given below and choose the correct answer.

   a) Make revolutionary advances in genetic disease therapy and treatment of cancer.

   b) Develop crops that are resistance to pests and diseases.

   i) Green Revolution ii) Gene Editing

   iii) Protein Synthesis iv) Co-dominance

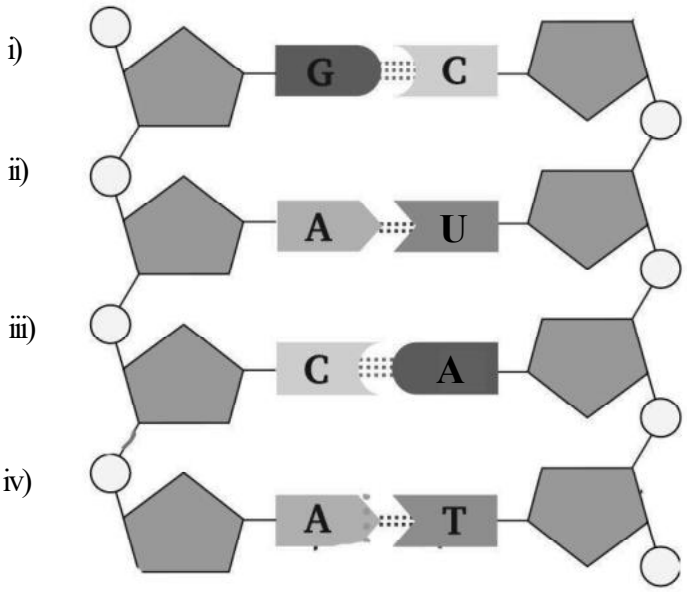
2. Match Coloumn I with Coloumn II and chosse the correct combination.

I	II
P Jennifer A Douodna	1. X-Ray Diffraction
Q Watson and Crick	2. Double helical model of DNA
R Rosalind Frunklin	3. CRISPR Cas 9

   a) P-3, Q-2, R-1 b) P-2, Q-3, R-1

   c) P-1, Q-2, R-3 d) P-2, Q-1, R-3

3. Analyse the illustration and choose the correct one.



a) i), ii)
b) iii), iv)
c) i), iv)
d) i), ii), iii), iv)

4. Which statement is not related to somatic chromosomes.

a) There are 22 pairs of somatic chromosomes in human.
b) One of these is inherited from the mother and the other from the father.
c) X and Y chromosomes are somatic chromosomes .
d) Control physical characteristics.

5. Select the statement related to genes

a) Specific sequence of nucleotides in DNA.
b) Basic building block of DNA
c) Made up of amino acids
d) Control metabolic activities

i) a) and b)
ii) a) and c)
iii) a) and d)
iv) i), ii), iii), iv)

6. Complete the table

	Number of strands	Type of Sugar Molecule	Nitrogen bases
DNA	2	(i)_____	A, C, G, T
RNA	1	Ribose	(ii)_____

7. Which one is related to transcription.

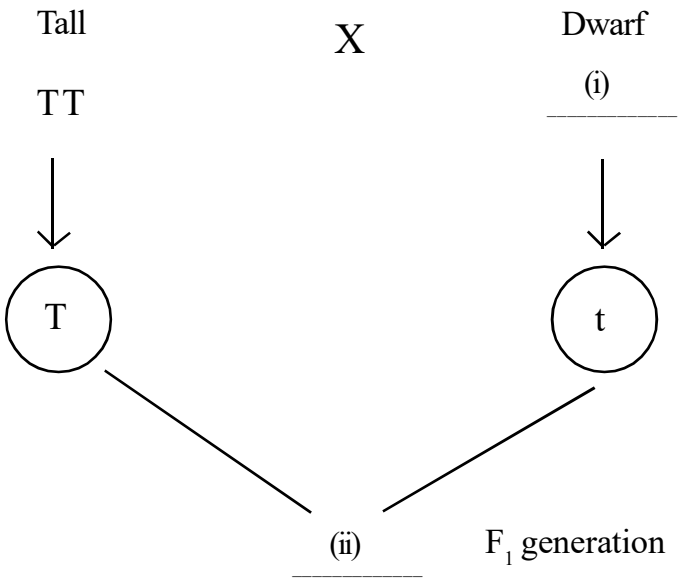
i) mRNA $\rightarrow$ DNA

ii) DNA $\rightarrow$ mRNA

iii) mRNA $\rightarrow$ Protein

iv) DNA $\rightarrow$ Protien

8. Complete the illustration



9. The hidden traits in an offspring is called .....

i) Dominant character ii) Co-dominance

iii) Reussive trait iv) Genotype

10. From the following select the statement which are not related to Monohybrid cross.

a) A pair of alleles in an organism does not influence the separation of another pair of alleles.

b) A dominant allele cannot fully hide the allele of the recessive trait.

c) The ratio of dominant to recessive traits in the offspring of the second generation is 3: 1.

d) When gametes are formed, the factors that determine trait gets separated with out mixing.

i) a and b ii) a and c

iii) b and d iv) c and d

11. Read the statement given below and choose the correct one.

Statement : Roan coat pattern found on some cattle and horses.

Reason : Both alleles exhibit their traits at the same time.

i) Both the statement and reason are incorrect

ii) Statement correct and reason incorrect

iii) Statement incorrect and reason are correct

iv) Both the statement and reason are correct

12. From the following which one is not related to meiosis.

a) Pairing of homologous chromosomes

b) Chromatids are not exchanged with each other

c) New traits are appeared

d) Causes the formation of gametes.

13. Which one is not a reason for mutation.

1) Chemical Substances 2) Radiation

3) Errors during DNA replication 4) Excercise

14. Analyse the following statement and choose the correct one.

i) F_1 has similarity with both the parent

ii) F_1 has no similarity with any of the parent character intermediate to them.

iii) F_1 has similarity with one of the parents.

a) i) Dominance ii) Incomplete dominance iii) Co-dominance

b) i) Incomplete Dominance ii) Dominance iii) Co-dominance

c) i) Co-dominance ii) Incomplete dominance iii) Dominance

d) i) Dominance ii) Co-dominance iii) Incomplete dominance

2 Mark Questions

1. Analyse the given illustration and write the answers for the questions.

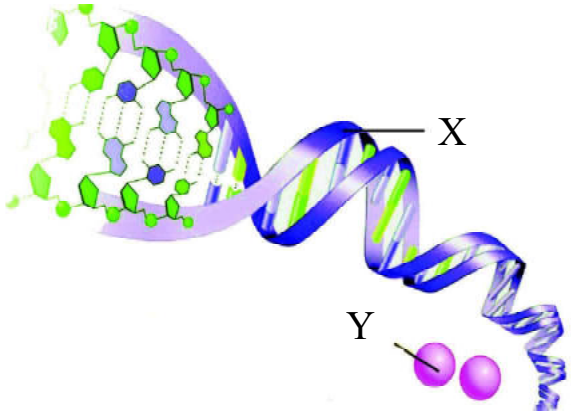


DNA

a) Molecules used to make strands of DNA.

b) Mode of nitrogen base pairing present in the rungs of DNA.

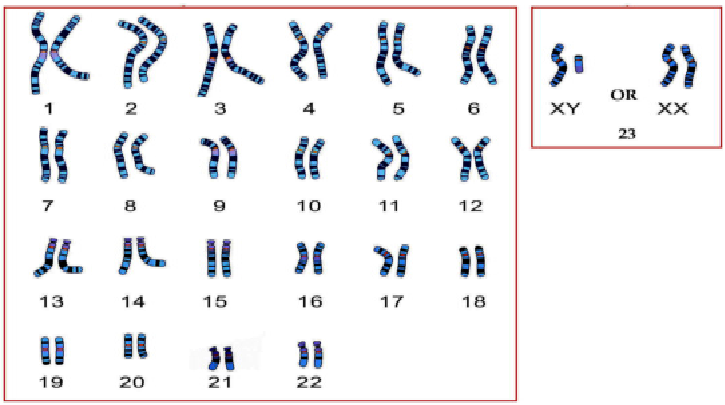
2. Observe the illustration and answer the following questions.



a) Identify X and Y

b) Explain the formation of nucleosome.

3. Observe the picture.



a) What is the function of SRY gene on the Y chromosome.

b) From the above, which one denotes the sex chromosome present in males.

4. Fill in the blanks.

	Genetic Constitution	Total no.of Chromosome	Total No.of Chromosome	Number and type of sex Chromosomes
Female	44+XX	46	i) ____	2, XX
Male	ii) ____	iii) ____	44 (22 pairs)	iv) ____

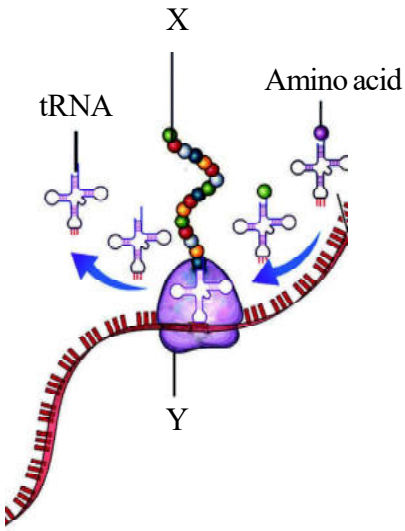
5. Observe the given picture and answer the following question.



a) Identify the picture labelled as 'X' and which is the nitrogen base present - only in it.

b) Name the Sugar molecule present in 'X' and also mention the number of carbon atom present in it.

6. Analyse the picture.



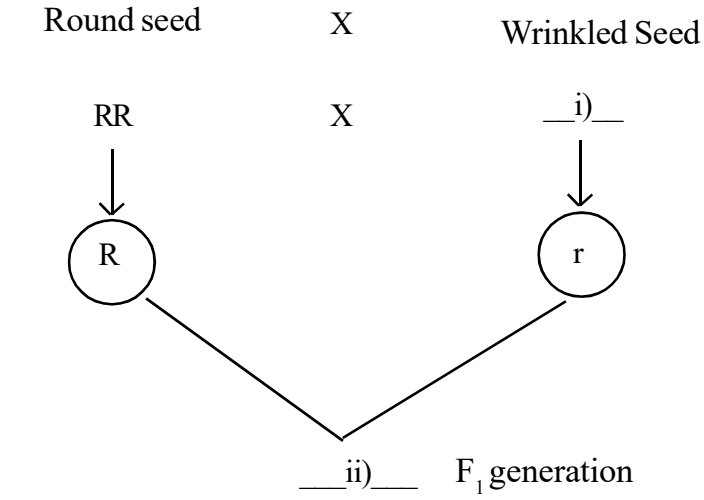
a) Identify the process mentioned here and also mention the place it occurs.

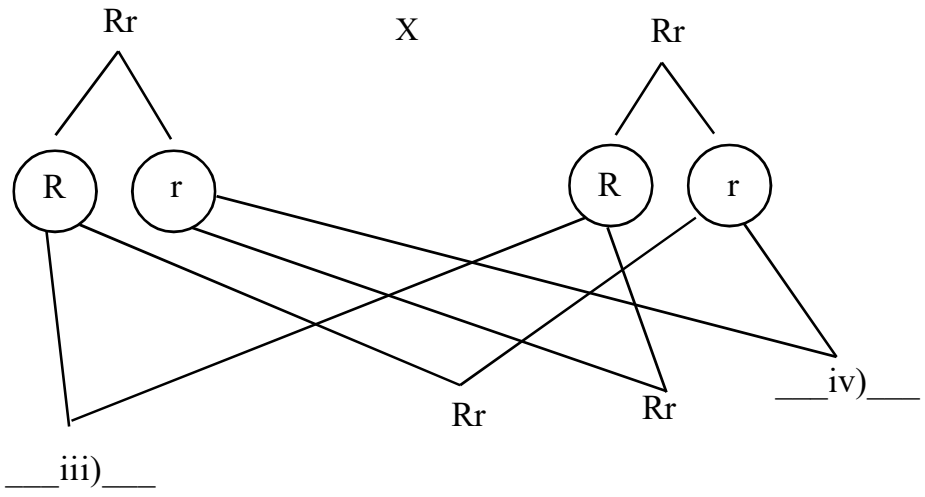
b) Identify X and Y.

7. One of the RNA involved in the protein synthesis is given below. Identify the othr RNA's and write down their function also.

tRNA $\rightarrow$ Carry specific amino acids to the ribosome for the protien synthesis.

8. Complete the illustration





9. The changes occured during the initial phase of meiosis is given below and answer the following questions.

a) Pairing of homologous chromosome

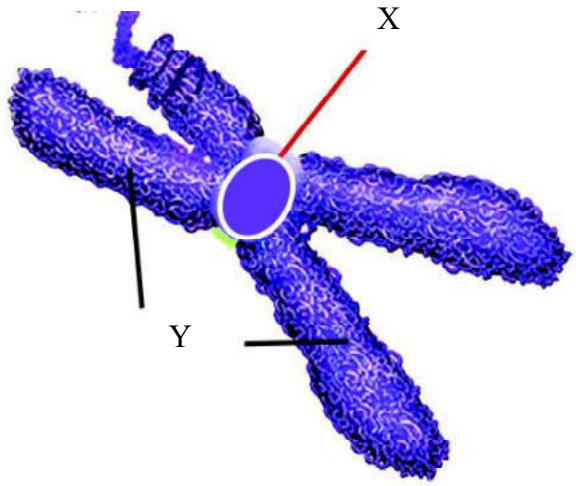
b) Chromatids break at the region of Chiasma.

c) Broken segments are exchanged with each other.

1) Mention the process mentioned above.

2) How this process leads to the formation of variations in offspring.

10. Observe the illustration and answer the following questions.



a) Identify the picture and how many pairs are present in human?

b) Identify X, Y

11. Some features of humans are given below. Analyse it and answer the following questions.

Indicators : X -ABO Blood group, Y Skin colour difference.

1) Write the genetic process behind X and Y.

2) Write the reason for X & Y?

3 Mark Questions

1. Illustrate the dihybrid cross between a tall plant with Axial flower (Dominant character, TTAA) and dwarf plant with terminal flower (Recessive character, ttaa)

a) Find out the genotype and phenotype of the parental plants.

b) What is the ratio of phenotype of F_2 generation?

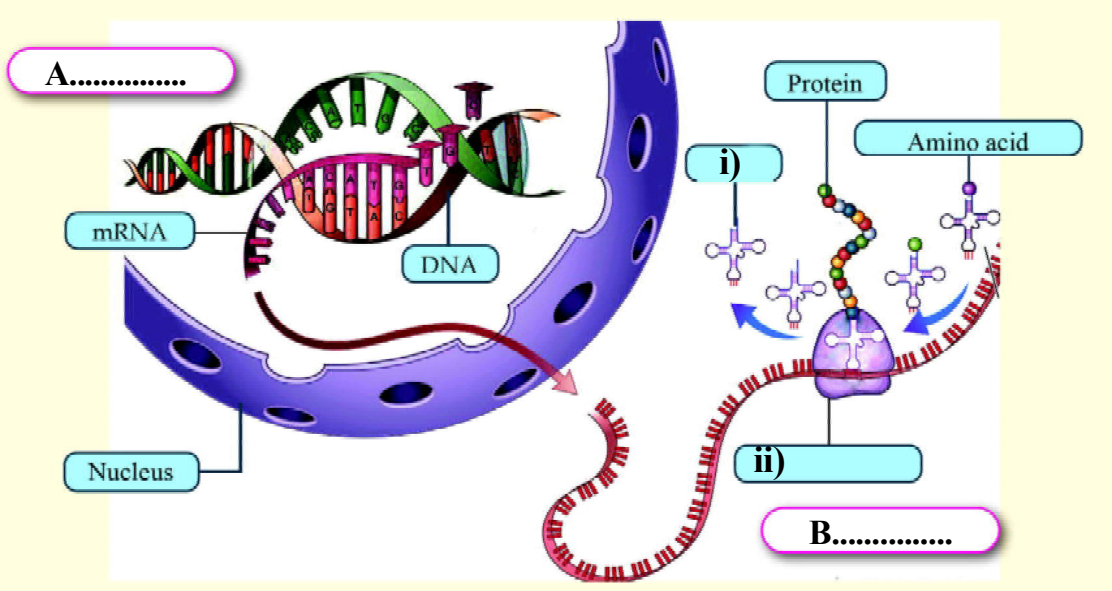
c) Find out Dominant and Recessive character of F_1 generation.

2. Complete the table.

Non Medelian Inheritance

Co-dominance	(a) _____	Incomplete Dominance	(b) _____
(c)_____	More than one gene control the trait.	(d) _____	Has more than two alleles
Roan coat	(e) _____	Pink coloured flower	(f) _____

3. Analyse the illustration and answer the questions given below



a) Identify the process?

b) Identify A and B?

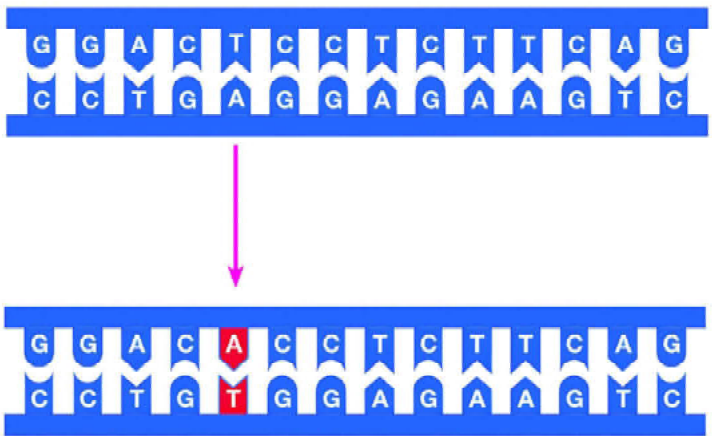
c) Write (i) and (ii)?

4. Complete the table

A	B	C
Co-dominance	___(a)___	Roan coat in animals
___(b)___	A dominant allele cannot fully hide the allele of the recessive trait.	___(c)___
Multiple Allelism	___(d)___	___(e)___
___(f)___	More than one gene control the trait	Difference in skin colour

4 Mark Questions

1. Observe the given picture and answer the following questions.



a) Identify and explain the process (1)

b) What is the reason behind this process? (2)

c) What is the importance of this process? (1)

2. Analyse the given indicator and answer the following questions.

Indicator : Proteins are synthesised as a result of the action of genes.

a) In which parts of the cell does this process takes place? (1)

b) Explain the phases of this process? (2)

c) Name the RNAs, which helps to form the bonds between amino acids in protein and the RNAs which carry amino acids to the ribosomes for protein synthesis. (1)

3. When a Tall plant with green colour seed (TTGG) hybridised with dwarf plant with yellow coloured (ttgg) seed, The F_1 generation become Tall plant with Green coloured seed (TtGg). Complete the table showing the F_2 generation when the F_1 generation is self pollinated.

i)

	TG	Tg	tG	tg
TG	TTGG	TTGg	(a)____	TtGg
Tg	(b)____	TTgg	TtGg	(c) _____
tG	TtGG	(d) ____	(e) ____	ttGg
tg	TtGg	Ttgg	ttGg	(f) ____

ii) Find out the phenotypic ratio of F_2 generation. (3)

Chapter 1

GENETICS OF LIFE

ANSWERS

1 Mark

1. ii) Gene Editing
2. a) P-3, Q-2, R-1
3. c) i), iv)
4. b) One of these is inherited from the mother and the other from the father
5. iii) a) and d) are correct
6. i) Deoxyribose ii) ACGU
7. ii) DNA $\rightarrow$ mRNA
8. i - tt, ii - Tt
9. iii) Recessive trait
10. ii) a and c
11. iv) Both the statement and reason are correct
12. b) Chromatids are not exchanged with each other
13. 4) Excercise
14. c) i) Co-dominance ii) Incomplete dominance iii) Dominance

2 Mark

1. a - Sugar, Phosphate

   b - A = T, G $\equiv$ C

2. a) X - DNA, Y - Histone protein

   b) Eight histone protein combained to form histone octamer. DNA is coiled around this is to form nucleosome.

3. a) Deveopment of testes in foetus.
   b) XY

4. i) 44
   ii) 44+XY
   iii) 46
   iv) -2, XY

5. a) X - RNA, Uracil
   b) Ribose Sugar, 5 Carbon atom

6. a) Translation, in Ribosome
   b) A - Protein B - Ribosome

7. mRNA - Carries messages for protein synthesis
   rRNA - Combines amino acids to make protein

8. i - rr, ii - Rr, iii - RR, iv - rr

9. 1) Crossing over
   2) In crossing over, the chromatids breaks and the broken segments are exchanged with each other, this cause recombination of alleles. This leads to variation

10. a) Chromosome, 23 pair
    b) X - Centromere Y - Chromatid

11. 1) X - Multiple Allelism
    Y - Polygenic inheritance
    2) X - The gene that determine blood group in human beings has more than two alleles. I^A, I^B and i determines the blood group.
    Y - more than one gene controlls the color of the skin. The action of these genes cause variation in the production of Melanin that causes difference in skin color.

3 Mark

1. a) Phenotype - Tall plant with Axial flowers
   dwarf plant with terminal flowers
   Genotype - TTAA, ttaa
   b) 9 : 3 : 3 : 1
   c) Dominant trait - Tall and Axial flowers
   Recessive trait - Dwarf and lateral flowers

2. a) Polygenic inheritance
   b) Multiple allelism
   c) Both alleles exhibit their trait at the same time
   d) A dominant alleles cannot fully hide the allele of recessive trait
   e) Skin color
   f) ABO blood group in human

3. a) Protein synthesis
   b) A - Transcription B - Translation
   c) i - tRNA, ii - Ribosome

4. a) Both alleles exhibit their trait at the same time
   b) Incomplete dominance
   c) Plants with pink flowers
   d) More than two alleles determines a character
   e) ABO blood group in human
   f) Polygenic inheritance

4 Mark

1. a) Mutation - Sudden heritable change in the genetic constitution of an organism
   b) Errors during DNA replication, exposure to certain chemicals, radiations
   c) Plays a major role in the process of evolution

2. a) Nucleus, Ribosome

b) Transcription - mRNA is formed from a specific nucleotide sequence in DNA
DNA → mRNA
Translation - Based on the messages in the mRNA, tRNA carry specific amino acids and rRNAs combine these amino acids to make protein

c) rRNA, tRNA

3. i) a - TtGG b - TTGg c - Ttgg
d - TtGg e - ttGG f - ttgg

ii) a) 3 : 3 : 1

Chapter 2

PATHS OF EVOLUTION

1 Mark Questions

1. Identify the statement related to Lamarckism

   1) Acquired character

   2) Food scarcity leads to variation

   3) Limitation of resources leads to competition among organisms.

   4) Evidences from the field of environmental science added to evolution theory.

   a) 1, 4 b) 2, 8

   c) 1, 2 d) 3, 4

2. Choose the correct one from the following related to Charles Darwin.

Year	Importance
a) 1809	1) Born in England
b) 1831	2) Voyage on the ship HMS Bedgle
c) 1836	3) Noticed the evolutionary studies of Alfred Russel Wallace.
d) 1858	4) Deeply Studies the specimens and observation he had collected
e) 1859	5) Presented 'On the origin of species'
f) 1882	6) Dead

   a) a-1, c-4, f-6 b) a-1, c-4, d-3

   c) a-1, e-4, f-6 d) a-1, c-5, d-3

3. Select the correct statement related to Darwinism.

i) Environment causes variation in organism

ii) Nature select the organism having favourable variation

iii) Giraffes with necks of varying length

iv) Giraffes of earlier periods possessed short necks.

a) ii, iii b) i, ii, iii

c) i, ii, iv c) i, iv

4. Identify the correct pairs from the option given below

Name of Scientist	Contribution
1) Thomas Malthus	P - Acquired character
2) Charles Darwin	Q - Increases in population leads to scarcity of resources.
3) Jean Bapstiste Lamarck	R - On the origin of species.

i) 1-P, 2-R, 3-Q ii) 1-R, 2-Q, 3-P

iii) 1-Q, 2-P, 3-R iv) 1-Q, 2-R, 3-P

5. Choose the correct statement regarding primates.

P - Thumb that cannot be opposed to other fingers

Q - Binocular and 3-D vision

R - Lives in groups

i) PQR Correct ii) Q and R Correct

iii) P and Q Correcct iv) R Correct

6. Evaluate the Statement given below.

a) Archea and Eukarya share a common ancestor; MRCA

b) Bacteria, Archea and Eukarya shares a common ancestor; LUCA

c) Protista, Plantae and Fungi shares a common ancestor; LUCA

a) a, b correct, C incorrect

b) a, b, c incorrect

c) a, b, c correct

d) a incorrect, b and c correct

7. While comparing the percentage of matching of the Chimpanzee's DNA sequences with the DNA sequences of other organism, choose the correct order of organism.

a) Monkey, Gorilla, Humans, Chimpanzee

b) Monkey, Chimpanzee, Gorilla, Humans

c) Monkey, Chimpanzee, Human, Gorilla

d) Monkey, Gorilla, Chimpanzee, Human

8. Evidnce given by fossils to evolution of life and its example is given below analyse it and choose the correct one.

Evidence : Evolution is a systematic process

Example : Dinosaur and mammoth

a) Evidence wrong and example correct

b) Evidence correct but example wrong

c) Both the evidence and example correct

d) Both the evidence and example wrong

9. The last organism in Hominoidea is human and the first organism is Gibbon. Select the correct order of organism from Gibbon to human.

A. Gibbon B. Oranguttan C. Gorilla D. Chimpanzee E. Human

a) A - B - C - D - E b) A - B - D - C - E

c) A - D - B - C - E d) A - D - C - B - E

10. Identify the correct pairs

Organisms in human evolutionary path	Cranial Capacity
a) Homo habilis	450 cm^3
b) Homo erectus	900 cm^3
c) Homo neanderthalensis	1450 cm^3
d) Homo sapiens	1550 cm^3

A) a, b

B) b, c

C) c, d

D) b, c

11. Choose the correct one.

Human	Characteristics
P. Astralopethecus	a) Bipedalism
Q. Homo habilis	b) Omni vores
R. Homo erectus	c) Began hunting

a) P-a, Q-b, R-C

b) P-a, Q-C, R-b

c) P-b, Q-c, R-a

d) P-b, Q-a, R-c

12. From the given option, select the parts present only in neurons.

a) Axon, Dendron, Cell body

b) Axon, Dendron, Synapse

c) Axon, Dendron, Ganglion

d) Axon, Dendron, Synaptic knob.

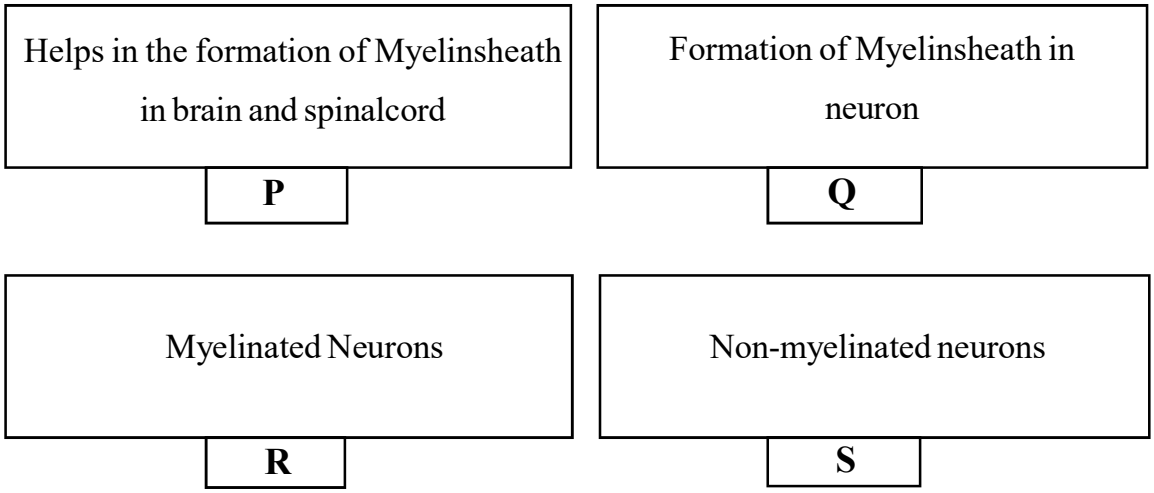
A) a, b

B) b, c

C) c, d

D) a, d

13. Analyse the information given in the boxes P, Q, R, S and identify the neuron parts associated with that.



A) P - Oligodendrocyte Q - Schwann cell
R - White matter S - Grey Matter

B) P - White matter Q - Grey Matter R - Oligodendrocyte S - Schwanncell

C) P - Schwanncell Q - Oligodendrocyte R - White matter S - Grey Matter

D) P - Oligodendrocyte Q - Gangliyon R - White matter S - Grey Matter

14. Choose the correct path of messages through neurons.

A) Brain $\rightarrow$ Sensory neurons $\rightarrow$ Receptor

B) Brain $\leftarrow$ Sensory neurons $\leftarrow$ Receptor

C) Brain $\rightarrow$ Motor neuron $\rightarrow$ Organs

D) Brain $\leftarrow$ Motor neuron $\leftarrow$ Organs

i) A, B ii) B, C iii) C, D iv) B, C

2 Mark Questions

1. Complex brain development played a crucial role in enabling humans to dominate nature and achieve cognitive and technological progress.

a) Which developed part of the brain is this?

b) How does man differ from other organism?

2. The brain is the center of the nervous system which controls and co-ordinates physiological function.

   a) What are the protective parts of the brain?

   b) What are the precautions for brain protection?

3. Withdraw the hand when unknowingly touching fire is a reflex action.

   a) What is reflex action?

   b) List the main parts in the pathway of impulses in a reflex arc in order.

4. Many functions like heartbeat and respiration occur in our body without conscious control.

   a) Which part of the nervous system controls involuntary actions.

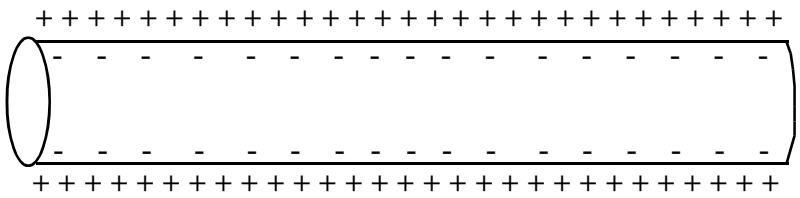
   b) Compare the functions of the sympathetic and parasympathetic nervous system?

5. For a nerve impulse to travel from one part of the body to another, it must pass through many neurons.

   a) Name the part where two neurons joined?

   b) What is the function of the synapse?

6. The illustration below indicates the transmission of impulses through a nerve cell. Analyse it and answer the questions.



   a) What change occurs in the state when stimulated.

   b) How does this change cause the transmission of impulses?

7. Observe the illustration and answer the questions.

Over production

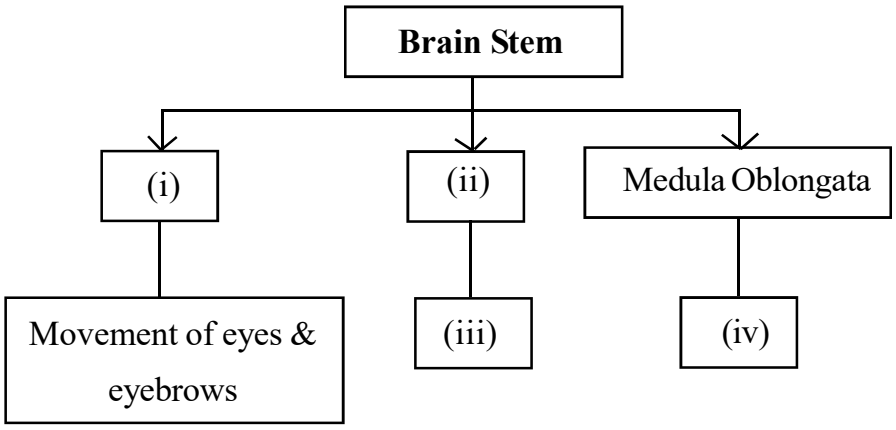
↓

Survival of existence

i) Which theory of evolution is indicated here?

ii) What is the importance of variation in the situation indicated in the illustration?

8. Complete the illustration

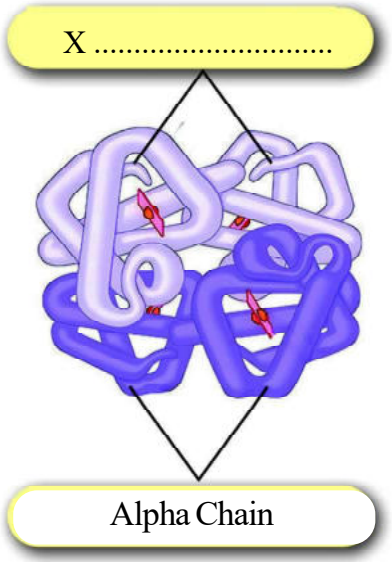


a) Write parts i) and ii)?

b) Write the function of iii) and iv)?

3 Mark Questions

1. Observe the illustration and answer the following questions

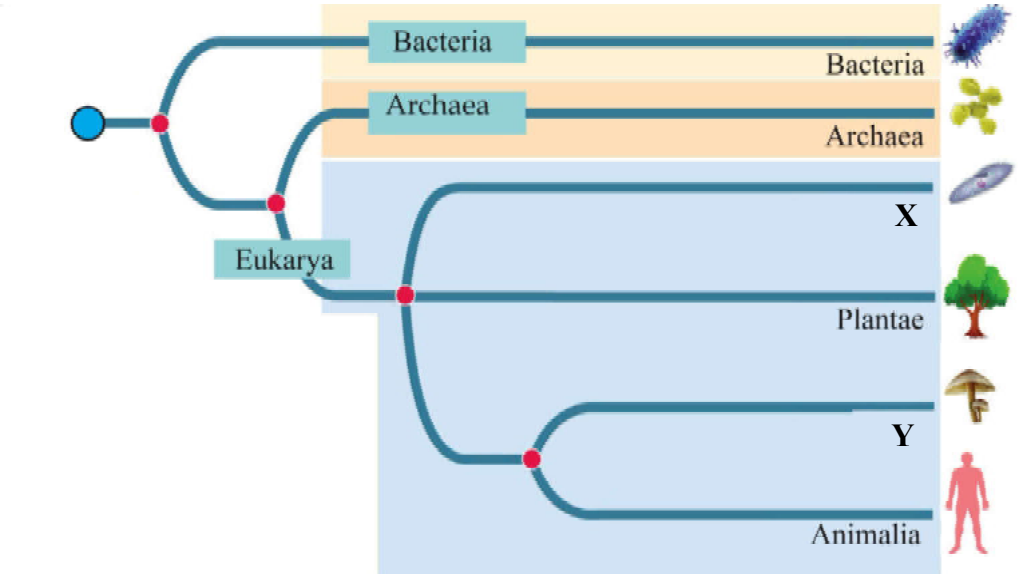


a) Identify the illustration, Label X?

b) Write the number of amino acids in 'X' do the animals Gorilla and Rat have compared to humans?

c) Which species have the most and least evolutionary relations with humans?

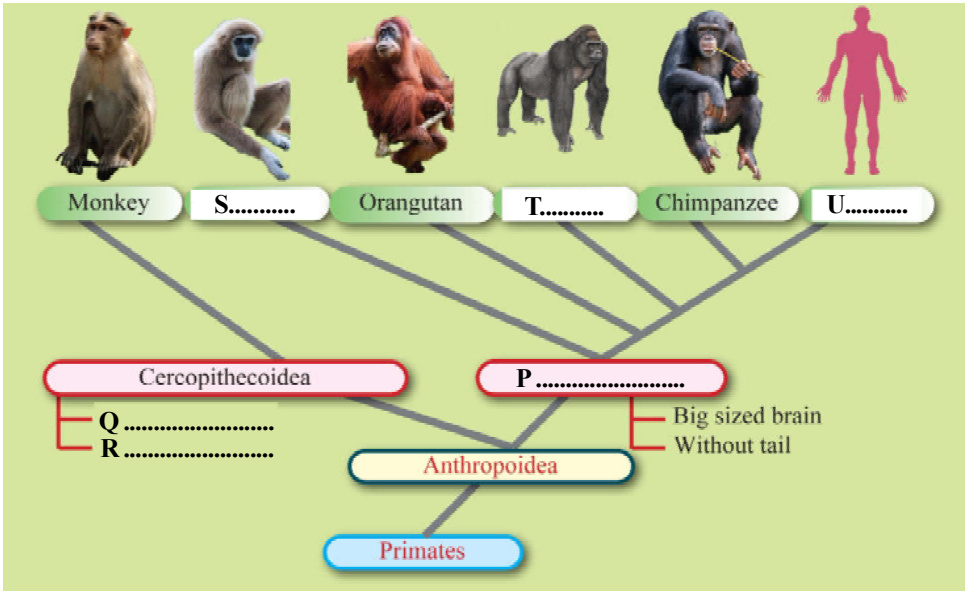
2. i) Observe the illustration and fill in the missing parts.



ii) What is the common ancestor shared by Archea and Eukarya? Who is the common ancestor of all of them?

iii) What is an evolutionary tree?

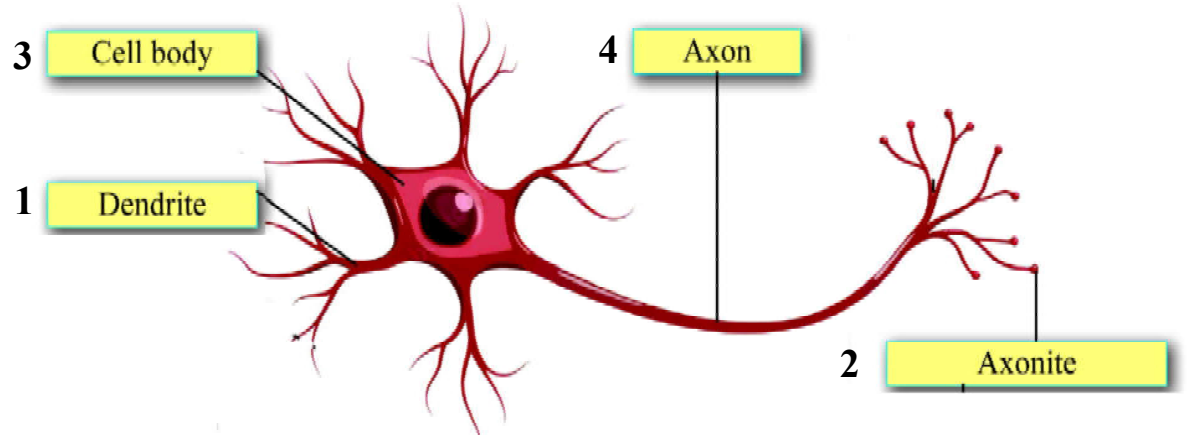
3. Observe the illustration and answer the following questions.



a) Name P, Q and R?

b) Write the names of the species S, T and U?

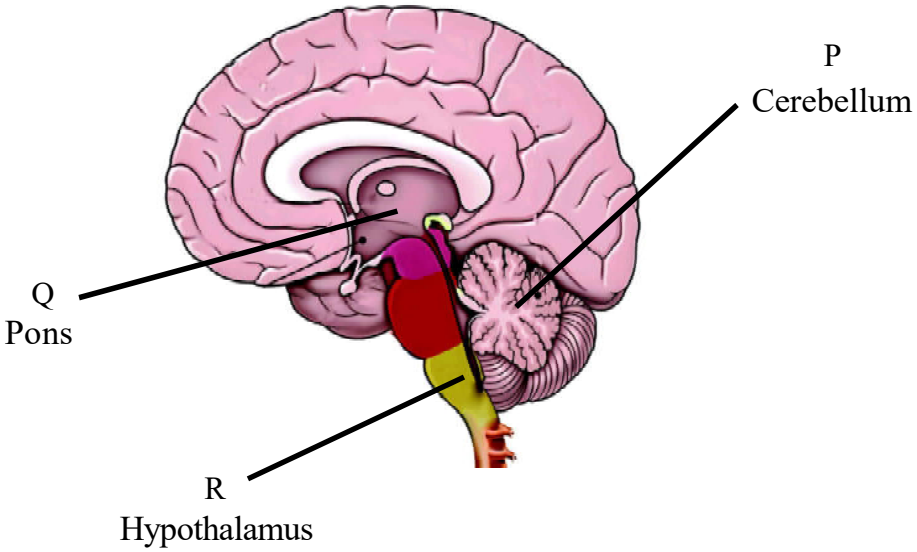
4. Redraw the diagram and correct the mistakes in the image.



a) Write an example for neurotransmitter?

b) Name the part of neuron which receives neurotransmitters?

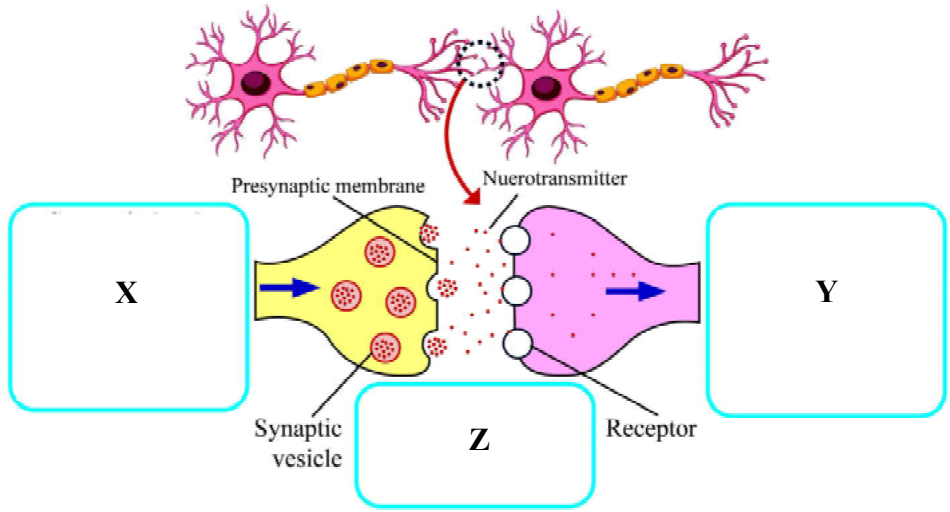
5. Redraw the image and answer the following question



Observe P, Q and R correct the labels if there are mistakes and write their functions?

4 Mark Questions

1. Observe the illustration and answer the following questions.

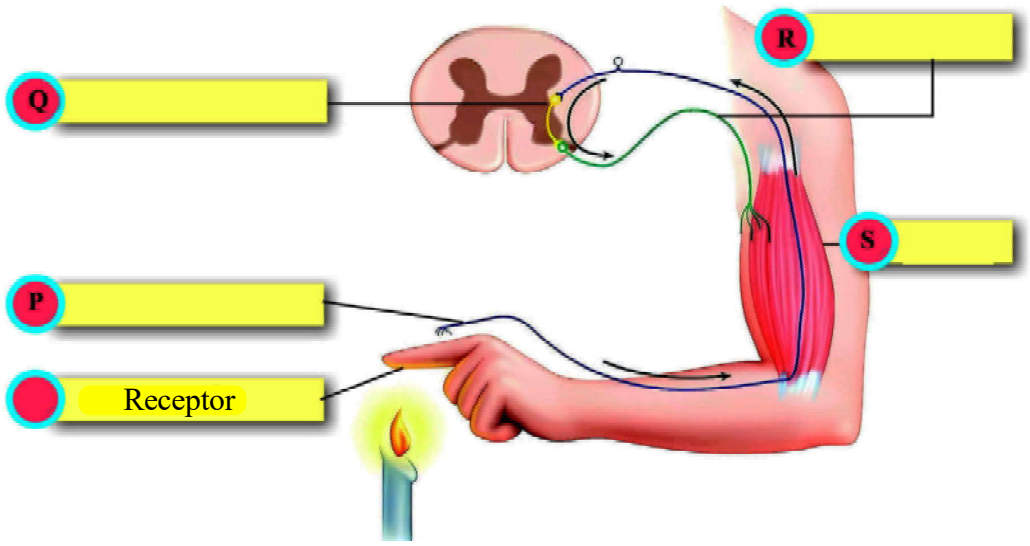


a) Identify the illustration and label X, Y, Z?

b) Where the post-synaptic layer containing receptor cells located in this diagram?

c) What is the function of the part illustrated in this diagram?

2. Analyse the illustration and answer the questions.



a) Name the process illustrated in this diagram and name its pathway?

b) Write P, Q, R and S?

c) What is the importance of this process?

Chapter 2

PATHS OF EVOLUTION

ANSWERS

1 Mark

1. c) 1, 2
2. b) a-1, c-4, d-3
3. b) i, ii, iii
4. iv) 1-Q, 2-R, 3-P
5. ii) Q and R Correct
6. a) a, b correct, c wrong
7. d) Monkey, Gorilla, Chimpanzee, Humans
8. b) Evidence correct but example wrong
9. a) A - B - C - D - E
10. D) b, c
11. b) P-a, Q-c, R-b
12. D) a, d
13. A) P - Oligodendrocyte Q - Schwann cell
    R - White matter, S - Grey matter
14. iv) B, C

2 Mark

1. a) Neo Cortex
   b) Human brain is more complex and developed than other organism. Advanced mutal process such as thinking, dicision making etc are made with the help of the synapse of neocortex.
2. a) Skull, meninges, cerebrospinal fluid
   b) Avoid accidents, wear helmet, seat belt etc

3. a) The reaction that occur spontaneously and invountarily
   b) Receptor → Sensory neuron → Interneuron → motor neuron → muscle

4. a) Autonomous nervous system
   b) Sympathetic system equips the body to respond during emergency situations. Parasympathetic system prepares the body to relax and perform routine function such as digestion.

5. a) Synapse
   b) Transmit the impulses to only one direction and increases the speed of the impulse.

6. a) The outer surface of the cell membrance is negatively charged and the inner surface is positively charged when stimulated.
   b) When stimulated, positive ions from outside the cell membrance enter the cell causing a temporary charge variation in that region.

7. i) Darwinism (Theory of natural selection)
   Charles Robert Darwin
   ii) Favourable variation overcome the struggle for existence. They become more productive.

8. a) i) Mid brain
   ii) Pones
   iii) Contros the rate of respiration
   or
   helps for the movement of eye muscles and facial muscles.
   iv) Controls involuntary actions

3 Mark

1. a) X - Beta chain
   b) Gorilla - 1, Rat - 31
   c) Most evolutionary relationship - Chimpansee
   Distance evolutionary relationship - Rat

2. i) X - Protista Y - Fungi
   ii) MRCA, LUCA
   iii) Evolutionary trees shows the evolutionary relationship between the organism

3. a) P - Hominoidea
      Q - Small sized brain
      R - Having tail
   b) S - Gibbon
      T - Gorilla
      U - Human

4. 1 - Dendron 2 - Synaptic knob
   a) Acetylcholine
   b) Dendrite

5. P - Cerebellum - Maintain equilibrium of the body
   Q - Hypothalamus - Maintain homeostasis
   R - Medulla oblongata - Controls involuntary actions

4 Mark

1. a) Synaps
      X - Synaptic knob, Y - Synaptic cleft
   b) Dendrite
   c) Transmit impulse to only one direction
      Increase the speed of the impulses

2. a) Reflex action, Reflex arc
   b) P - Semory neuron
      Q - Inter neuron
      R - Motor neuron
      S - Muscle
   c) These actions are the reaction that occur spontaneously and involuntarily in response to stimuli.

Chapter 3

BEHIND THE SENSATION

1 Mark Questions

1. Analyse the following statement and find the correct answers.

   i) The conjunctiva is the protective covering of the front part of the eyes, which includes the cornea and eyelids.

   ii) Body temperature increases if there is an infection

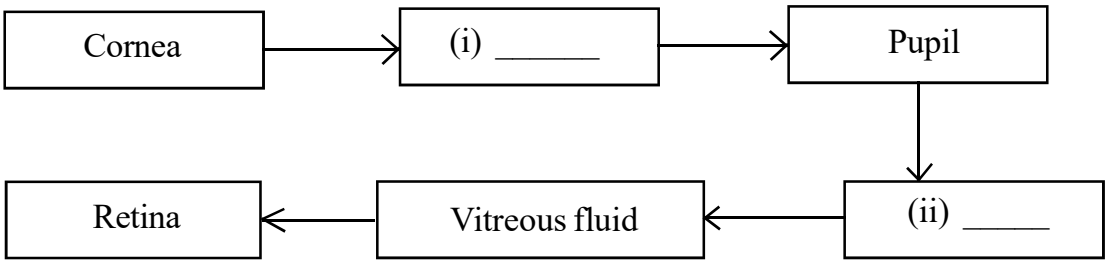
   iii) Sensation like touch, pain, heat and pressure are special sensation

   iv) Stimuli are the situation that causes responses in living beings.

   a) i, ii are correct b) i, ii, iii are correct

   c) ii, iv are correct d) All are correct

2. Complete the following illustration



3. Which part transfers messages from the bipolar cell layer to the optic nerve in the eye?

   a) Layer of photoreceptors b) Ganglion layer

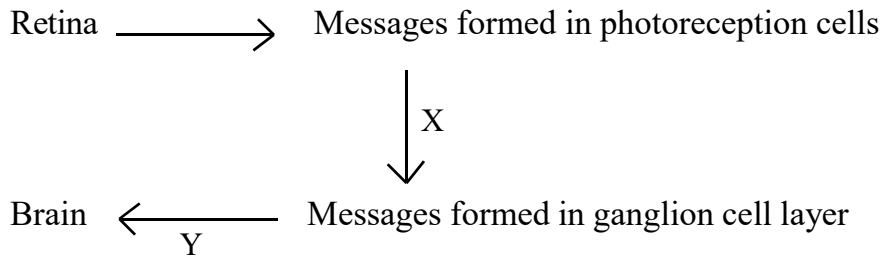
   c) Retina d) Monopolar cell layer

4. Correct if there is an error.

a) The central aperture of the Iris is known as the cornea.

b) The size of the pupil is 2 to 3 millimeters.

5. Observe the illustration given below and mark X, Y.



6. Which of the folowing are bipolar cells?

a) Monopolar cells

b) On bipolar cells

c) Tripolar cells

d) Off bipolar cells

7. Analyse the following statement and reason and write the answer.

Statement : Aqueous humor is a fluid that is formed from the blood and is reabsorbed back in to the blood.

Reason : Therefore the shape of the eye ball is maintained.

a) The statement and the reason are correct

b) The statement is correct, the reason is wrong

c) The statement is wrong and reason is correct

d) The statement and the reason are wrong

8. Analyse the following statements and write the answer.

Statement A : The blind spot is the part where optic nerve begins.

Statement B : Photoreceptor cells are not seen in the blind spot

a) A and B are correct

b) A is correct and B is wrong

c) A is wrong B is correct

d) A and B are wrong

9. When the circular muscle contract in bright light the size of the pupil,

a) Increases b) Decreases

c) Does not change d) Increases and decreases

10. When radial muscle contract in dimlight, the size of the pupil,

a) Decreases b) Increaes

c) Increases and decreases d) Does not change

11. Which of the following is not a part of the lens?

a) Capsule b) Vestibule

c) Lens fibers d) Epithelium

12. Analyse the following statement and choose the correct one.

Statement

The ability of the eye to precisely focus the image of near and distant object or retina is called persistence of vision.

Reason

This is achieved by changing the curvature of the lens through the action of ciliary muscles.

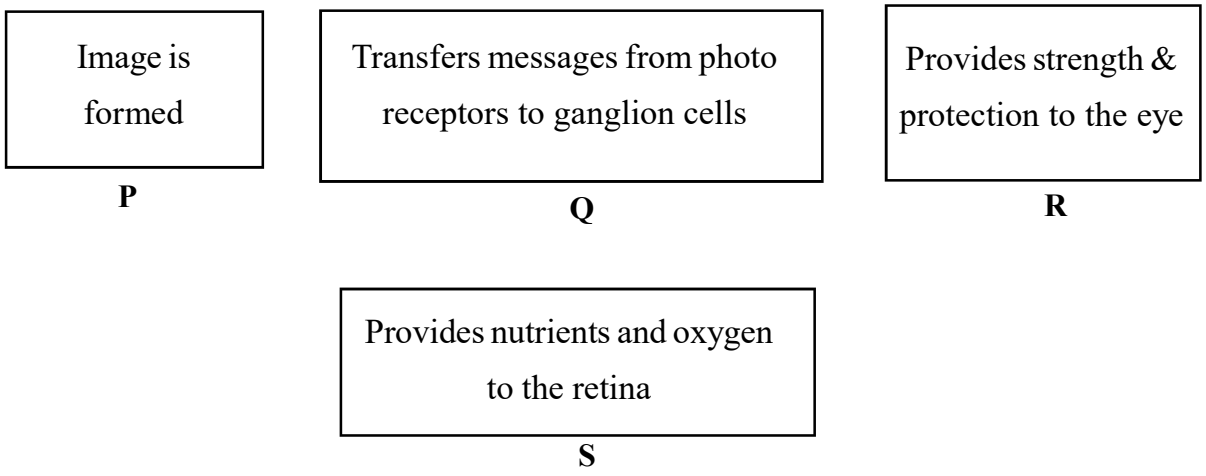
a) The statement and the reason are wrong

b) The statement is correct, reason is wrong

c) The statement is wrong, the reason is correct

d) The statment and reason are correct

13. Analyse the information in the boxes indicated as P, Q, R, S and identify the corresponding parts of the eye and choose the correct answers.



a) P - Choroid, Q - Retina, R - Bipolar cell layer, S - Sclera

b) P - Retina Q - Sclera, R - Choroid, S - Bipolar cell layer

c) P - Retina, Q - Bipolar cell layer, R - Sclera, S - Choroid

d) P - Choroid, Q - Bipolar cell layer, R - Retina, S - Sclera

14. Choose the correct statement from the following.

"When looking at a distant object"

a) Ciliary muscles relaxed

b) Curvature of the lens increases

c) Ligaments are stretched

d) Ciliary muscles contract

2 Mark Questions

1. Analyse the statement and reason and choose the correct answer

Statement : Tears protect the eye from infections

Reason : Tears contain Hydrochloric acid

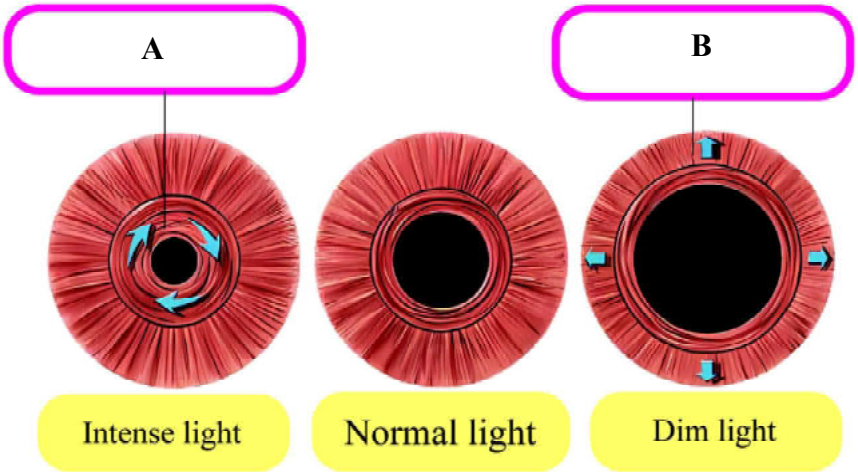
a) Statement correct, Reason wrong

b) Statement and reason are wrong

c) Statement and reason are correct

d) Statement is wrong, reason is correct

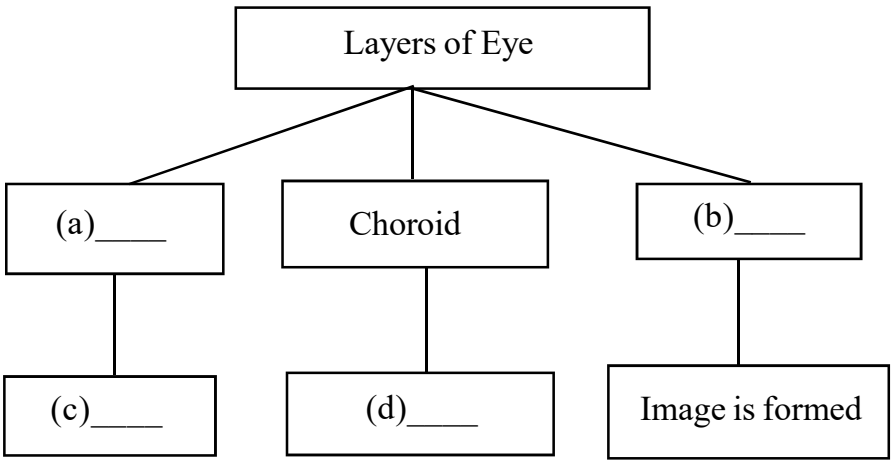
2. Observe the illustration of the muscles related to the regulation of the pupil size and answer the questions.



a) Which muscle do A and B indicate?

b) How do these muscles function according to the variations in light intensity.

3. Complete the following flow chart.



4. Choose the correct statements about the chambers of the eye

a) The upper and lower chambers are filled with a fluid called perilymph

b) The chamber between the cornea and the lens is called the aqueous chamber

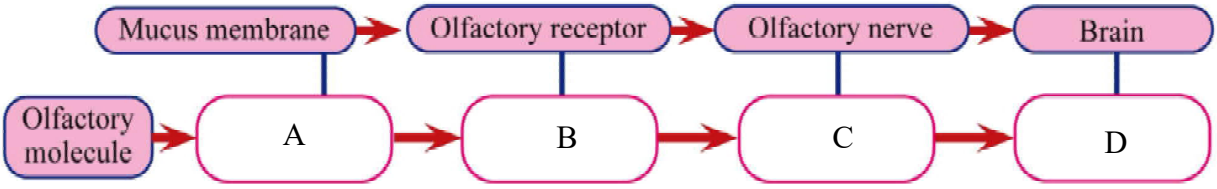
c) Aqueous humor is a fluid that is formed from the blood and is reabsorbed back in to the blood

d) Vitreous humor is a water like fluid

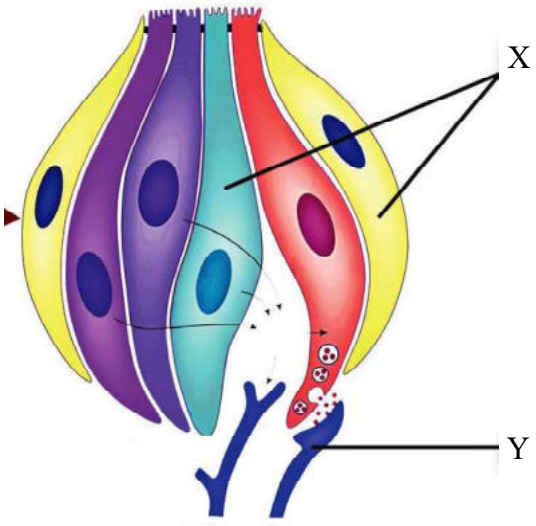
e) The lens and cornea receive oxygen and nutrients through aqueous humor

f) The vitreous chamber is located between the lens and the retina

5. Complete the flow chart given below



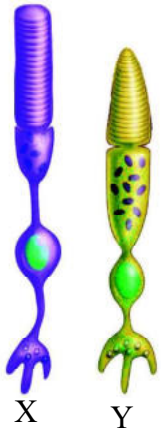
6. Analyse the illustration given below and answer the following question.



a) Identify the illustration and in which organ it is found.

b) Identify X and Y?

7. Observe the illustration and answer the questions given below



a) Identify X, Y

b) Identify the components in them.

8. Through the process calle Binocular fusion the two images we see are compared and merged in to one

a) Where does this process take place?

b) What is its importance of this process?

9. Analyse the following treatment and causes, and identify the eye diseases.

1) Concave lens spectacules | contact lens surgery | elongated eye ball :

a) .....

2) Cylindrical lens | Irregular cornea | lens :

b) .....

3) Antibiotics are used | infections of the conjunctiva :

c) .....

4) Food item containing vitamin A | Capacity of cornea decreases due to prolonged deficiency of vitamin A.

d) .....

10. Which is not a characteristics of the ear canal?

1) Helps to identify which direction the sound is coming from

2) Protects the ear drum from foreign objects

3) Ear wax is present

4) The diameter of the ear drum is 9-10mm

5) Oscicles are present

6) Air pressure is equalized

11. "Women are likely to be affected by color blindness"

Find whether the statement is correct or wrong and write the reason.

12. Visual Cortex and Binocular fusion are two term related to vision.

Compare them.

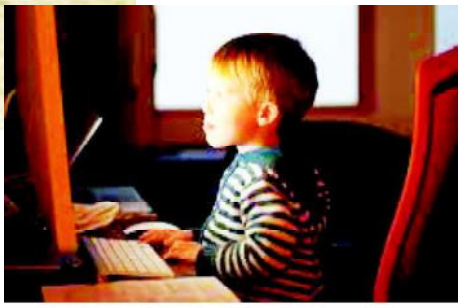
13. All living being have sense organ to perceive the environment.

Write two sense organs of insects and their function.

14.

Children below the age of five should spend only an hour or less on digital devices. Those who are below one year should not use such devices.

-World Health Organisation (WHO)



Indicators :

* Screen time * Sleep * Personal relationship * Physical and mental health * Consequences

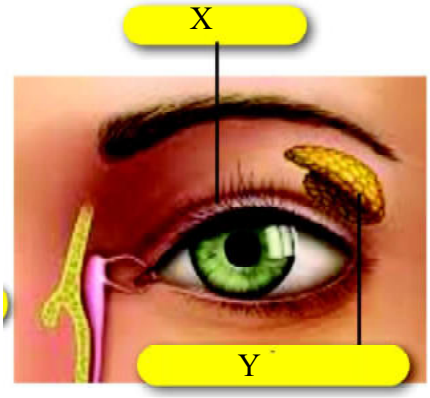
Analyse the above statement based on the indicators and prepare a note on digital usage of adolescents.

3 Mark Questions

1. Complete the table.

Sense Organ	Receptor	Stimulus
Eye	Photo receptors	(i) _____
Ear	(ii) _____	(iii) _____
Nose	(iv) _____	Odour
Tongue	(v) _____	(vi) _____

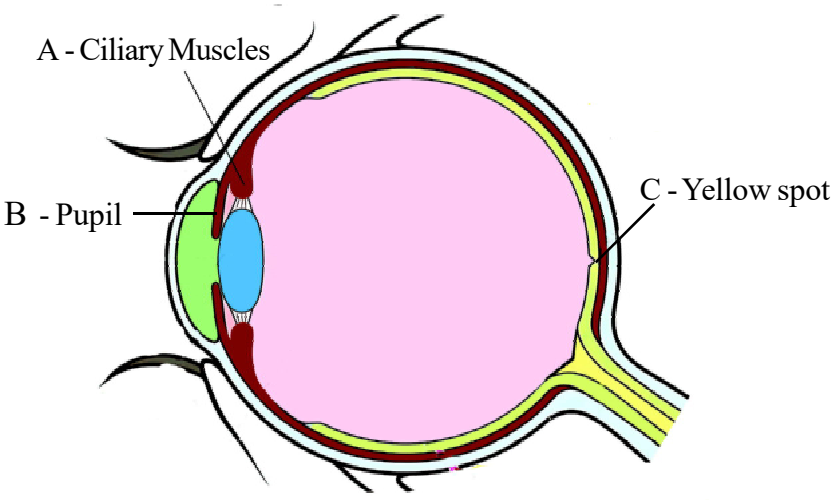
2. Based on the illustration of the anterior part of eye, answer the questions given below.



a) Identify X, Y

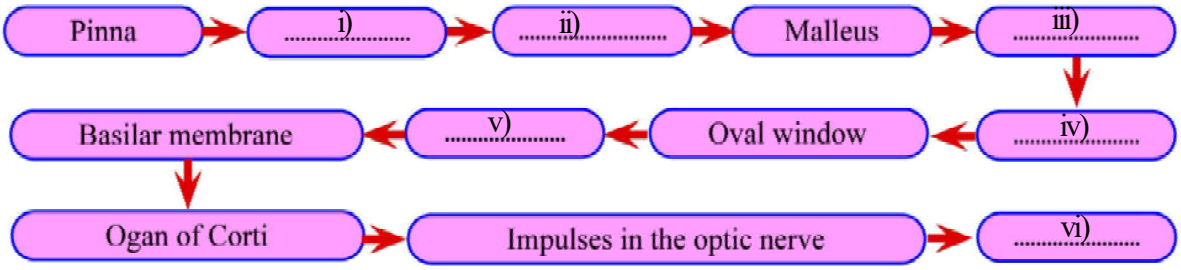
b) What are the functions of Y?

3. Redraw the diagram given below and write the answers to the questions given below.



a) If there any mistake in the labelling of A, B, C correct them and write the functios of A and C.

4. Complete the following flow chart



5. Complete the table

Part	Function
Pinna	(a) _____
(b) _____	Directs sound waves to ear drum
Ear drum	(c)_____
(d) _____	Causes sound waves to vibrate
(e) _____	Directs sound waves to the inner ear
Auditory nerve	(f) _____

6. Analyse the image given below ad write down the answers.

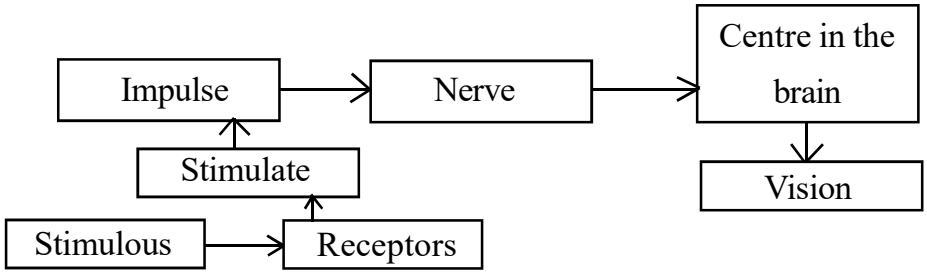


i) Identify the image, name the part which helps to maintain body balancing?

ii) Name two movement which helps to hear?

iii) Name the Fluid present in innear ear?

7. Recreate the illustration as given by including the sense organ eye.



8. Analyse the given word pairs below and find out the relationship between them.

a) Retinal - Night blindness

b) Umami - Taste buds

c) Cone Cells - Color blindness

9. Describe the relationship of the process with the occurance of the sensation of smell.

Process	Part
Substances dissolves in	a)
Stimulating receptors	b)
Nerve that carry impulses to brain	c)

10. Observe the indicators given below and rearange the following part appropriately.

Parts : Ear Oscicles, Semi circular canal,
Auditory nerve, Pinna, Auditory canal,
tympanum vestibule, cochlea, vestibular nerve.

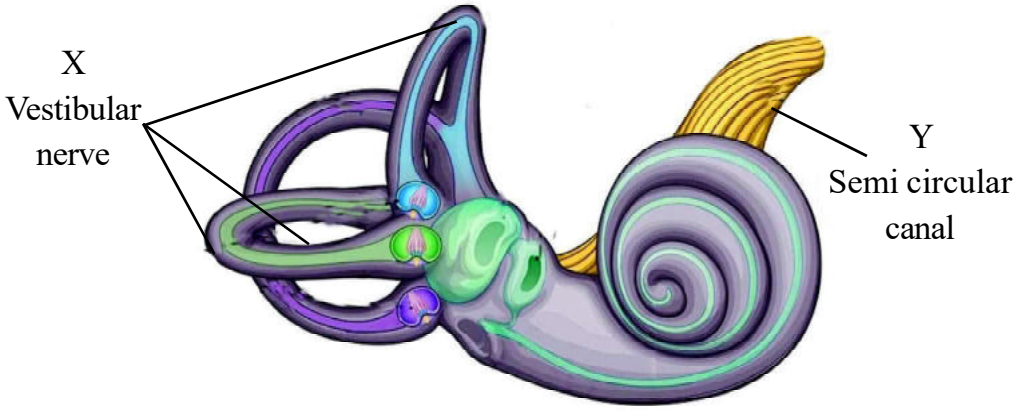
Indicators:

* Outer ear

* Middle ear

* Inner ear

11. Redraw the diagram and correctly label the parts of the vestibular system, that function in maintaining body balance and write down its function.



4 Mark Questions

1.

Eye disease	Reason	Treatment
(a)_____	Eye lens becomes opaque	(b) _____
Short sightedness	(c) _____	Spectacles with Concave lens
(d) _____	Infections in the Conjunctiva	Use of antibiotic eye drops with prevent the entry of pathogen.
Long sightedness	(e) _____	Spectacles with Convex lens
(f) _____	Due to prolonged deficiency of Vitamin A Cornea become opaque	(g) _____
Astigmatism	h) _____	Cylindrical lens

2.



A B C

i) Complete the table by identifying A, B and C based on the changes in Pupil.

A	B	C
Circular Muscles	Circular muscles Radial muscles	Radial muscles
i)	ii)	iii) .....

ii) Size of pupil (A)_____ (B) _____ (C) _____

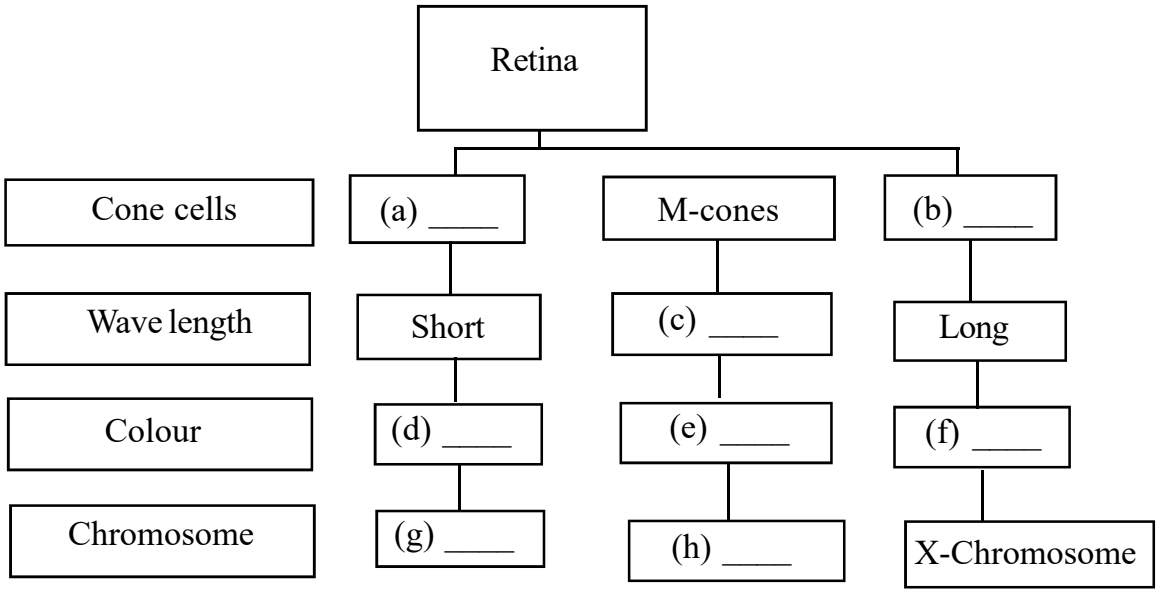
iii) Name the pigment present in Iris? Write down it's functions?

3. Analyse the statement given below and write down the relationship between photo receptors and vision.

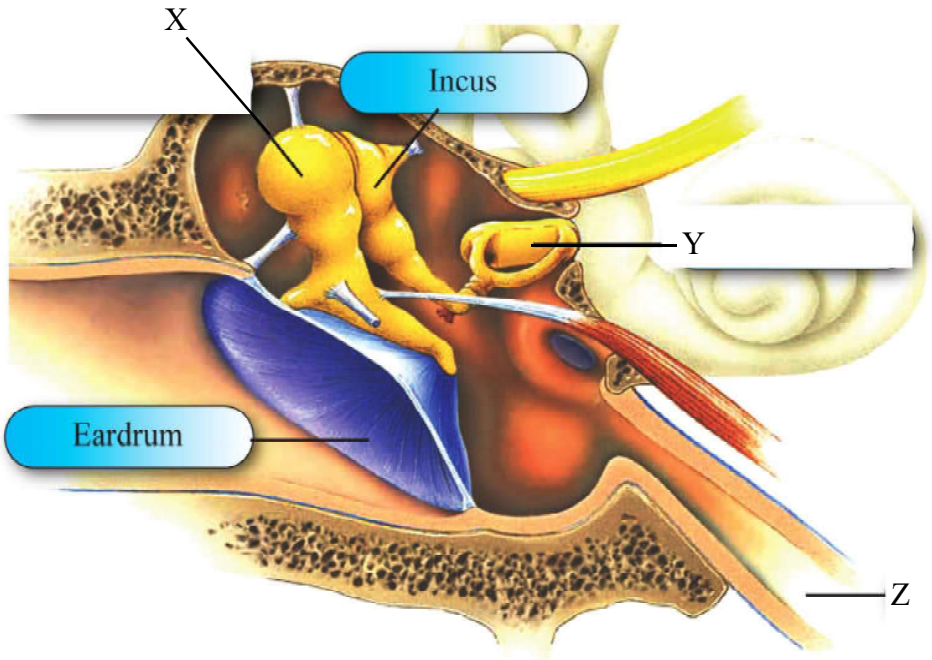
Statement : Glutamate acts as the primary neuro transmitter in photo receptors variation in glutamate production are responsible for the perception of darkness and light.

	Dark	Light
Glutamate	(a) _____	(b) _____
On bipolar cells	(c) _____	(d) _____
Off bipolar cells	(e) _____	(f) _____
	(g) _____ In the absence of light form impulses	(h) _____ Presence of light form impulses

4. Complete the flow chart given below



5. Analyse the illustration and answer the questions

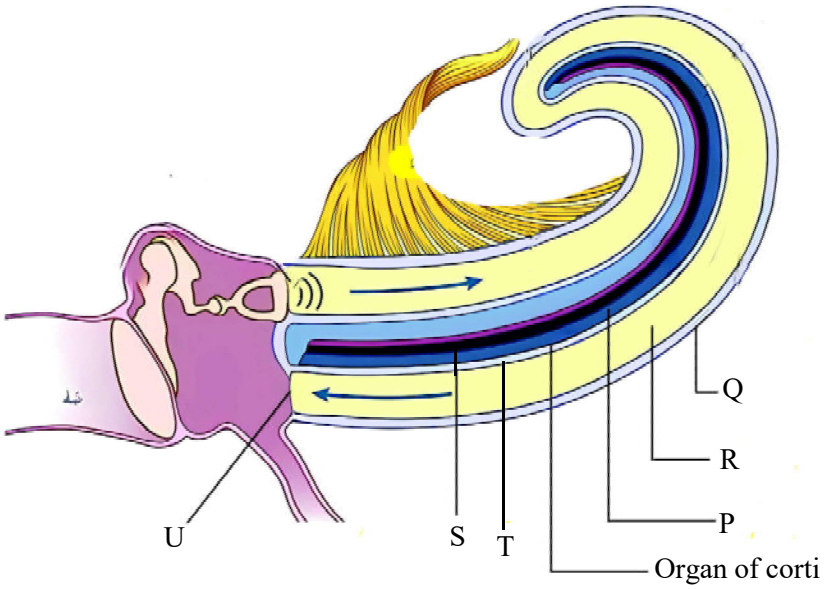


a) Identify the image, write down X, Y, Z?

b) Write down the pecularities of ear drum?

c) Write down the function of 'Z'?

6. a) Analyse the illustration given below and find out P, Q, R, S, T and U?



b) Write down the location of organ of corti in the above illustration? What is its function?

7. The sensory diversity in organism is given below in the table. Analyse and answer the questions given below.

Organism	Sensory organ
1) Euglena	Eye spot
2) Insects	Ommatidia
3) Bat	___(a)___
4) Snake	___(b)___

1) Answer a and b?

2) Compare the sensory capacity of dog and Hawk?

3) Which organism posess antenna as sensory organ? Write down its function?

Chapter 3

BEHIND THE SENSATION

ANSWERS

1 Mark

1. c) ii, iv are correct
2. i) Aques fluid ii) Lens
3. b) Ganglion cell layer
4. The aperture seen at the center of iris is called pupil
5. X - Bipolar cell layer Y - Optic nerve
6. b) On bipolar cells
   d) Off bipolar cells
7. b) The statement is correct, the reason is wrong
8. a) A and B are correct
9. b) Decreases
10. b) Increaes
11. b) Vestibule
12. c) The statement is wrong, the reason is correct
13. c) P - Retina, Q - Bipolar cell layer, R - Sclera, S - Choroid
14. a and c are correct

2 Mark

1. a) Statement correct, Reason wrong
2. a) A - Circular muscle B - Radial muscle
   b) In intense light circular muscle contract, in dim light radial muscle contact.
3. a) Sclera b) Retina c) Provides firmness and protection
   d) Provides oxygen and nutrients to the inner layer of retina.

4. b) The chamber between the cornea and the lens is called the aqueous chamber

c) Aqueous humor is a fluid that is formed from the blood and is reabsorbed back in to the blood

e) The lens and cornea receive oxygen and nutrients through aqueous humor

f) The vitreous chamber is located between the lens and the retina

5. A - Dissolves in nucleus

B - Get stimulated and generate impulses

C - Carries impulses

D - Sense of smell

6. a) Taste buds, tongue

b) X - Chemo receptors

Y - Neurons

7. a) X - Rod cells

Y - Cone cells

b) Opsin, Retinal

8. a) Visual cortex

b) Get 3D vision/helps to determine the differences between two images/enables to perceive depth.

9. 1) Short sightedness

2) Astigmatism

3) Conjunctivities

4) Xerophthalmia

10. 2) Protect the tympanum from foreign particles

3) Ear wax present

11. Statement is correct.

Reason : The gene responsible for the production of pigments in cone cells which are sensitive to green and red are found in the X-chromosome. XX Chromosome present

in women, so the possibility of colorblindness is less in women.

In the case of male, there is only one X-Chromosome. So the possibility of colorblindness is more in men.

12. There will be two slightly different images form in retina. Through optic nerve these images reached to visual cortex in brain. The brain compares these images and combines them. This process is called binocular fusion.

13. The compound eye made up of ommatidia helps for vision and movement. Antena helps to detect smell and touch.

14. Short note

3 Mark

1. i) light
   ii) Auditary receptors
   iii) Sound
   iv) Olfactory receptors
   v) Chemoreceptors
   vi) Taste

2. a) X - Eye hids Y - lacrimal gland
   b) • Keeping the surface of the eye moist
      • Eliminating waste
      • Protect from infection

3. A - Ciliary muscles
   B - Iris
   C - Yellow spot
   A - Function - Adjust the curvature of lens
   C - Function - Clear vision is possible because much photoreceptor cells are present.

4. i - Audtary canal ii) tympanum iii) Incus
iv) Stapes v) Cochlea vi) Brain

5. a) Direct the sound waves in to auditary canal
b) Auditary canal
c) Vibrate according to the sound waves
d) Ear oscicles
e) Oval window
f) Transmit impulses to brain

6. i) Internal ear, vestibular system
ii) Rotational movement and perpendicular movement
iii) Endolymph, Perilymph

7. Light → Photo receptors → Stimulous → Impulses → Optic nerve → Cerebrum → Vision

8. a) Retinal is a component of photo receptor cells it is derived from Vitamin A. Nightblindness occurs due to deficiency of vitamin A.
b) One of the main taste is Umami, Taste buds help to distinguish this.
c) Cone cells helps to distinguish colors. The defects in cone cells leads to color blindness.

9. a) Mucus b) Olfactory receptors c) Olfactory nerves

10. Outer ear - Pinna, ear canal
Middle ear - Ear drum, ear oscicles
Internal ear - Semi circular canal, cochlea, vestibule, vestibular nerve.

11. X - Semi circular canal - Maintain body balance
Y - Vestibular nerve - Carries impulses to brain

4 Mark

1. a) Cataract
   b) Surgery
   c) Enlarged eye ball
   d) Conjunctivitis
   e) Shortened eye ball
   f) Xerophthalmia
   g) Eat Vitamin A containing foods
   h) Cylindrical lens

2

	A	B	C
i)	Contract	ii) Retain its original position	iii) Contract
ii)	A) decreasing	B) No change	C) Increasing
iii)	Melanin - Absorbs	Ultraviolet rays	

3. a) Producing
   b) Not producing
   c) Inactivating
   d) Activating
   e) Stimulating
   f) Not working
   g) Off bipolar
   h) On bipolar

4. a) S Cones
   b) L Cones
   c) Medium
   d) Blue
   e) Green
   f) Red
   g) 7
   h) X Chromosome

5. a) Middle ear
   X - malleus
   Y - Stapes
   Z - Eustachian canal
   b) 9 - 10mm diameter
   0.1mm thickness
   c) Helps to balance the pressure on both sides of typanum

6. a) P - Middle ear

Q - Cochlea

R - Perilymph

S - Endolymph

T - Basilar membrance

U - Round window

b) The organ of corti, where the auditory receptors are present is stimulated in the basilar membrance. The vibration that reach the hair cells present here generate impulses.

7. 1) a) Ear/Ecolocation organ

b) Jacobson's organ

2) In dog highly sensitive olfactory receptors are found. In Hawk, eyes with high vision systems for long distance vision and detecting ultraviolet radiation.

3) Insects, it helps to detect smell and touch.

•

Chapter 4

CHEMORECEPTION IN ORGANISMS

1 Mark Questions

1. Find and write the correct statement

   i) Cytokinins stimulate cell division and differentiation in germinating seeds.

   ii) Auxins stimulate the growth of fruits and seeds

   iii) Gibberlings causes the shedding of mature leaves abd fruits

   iv) Abscisic acid maintains the dormancy of seeds until favourable condition arise

   a) i), ii) are correct b) i), iv) are correct

   c) ii), iii) are correct d) iii), iv) are correct

2. Synthetic plant hormones used in agricultural sector and their uses are listed. Find out the incorrect one from this?

A	B
i) Ethyline	iv) Used to ripen fruits simultaneously
ii) Auxin	v) Used for stem elongation in sugar cane
iii) Gibberlin	iii) Used to promote root growth in stem

   a) ii) - v), iii) - iv) b) i) - iv), ii) - v)

   b) ii) - iv), iii) - iv) c) i) - iv), iii) - vi)

3. What are the special cells on which hormones can act?

a) Nerve cells
b) Target cells
c) Blood cells
d) Glandular cells

4. Analyze the statement and reason and find the correct answer.

Statement A : Target cells have receptors

Statement B : Receptors help to recognize hormones

a) Both statements are correct
b) Statement A correct, B false
c) Statement B correct, A false
d) Both statements are false

5. Analyse the statement and reason and find the correct one.

Statement : Always grows towards the direction light

Reason : When sunlight falls, Auxin is equally distributed to all parts of the cell

a) Statement is correct, reason is false
b) Both statement and reason are correct
c) Statement is false, reason is correct
d) Both statement and reason are false

6. Analyse the statement and reason, and find the correct one.

Statement : Cytokinins help seeds to germinate

Reason : Cytokinins stimulate cell division and differentiation in seeds

a) Statement is false, reason is correct
b) Both statement and reason are false
c) Both statement and reason are correct
d) Statement is correct reason is false

7. Analyse the given statement and reason given below. And write the correct answer.

Statement : Calcitonin is produced when the level of calcium in the blood increases.

Reason : Calcitonin regulates the calcium level by reducing the mixing of calcium from bones in to the blood.

a) Both statement and reason are correct

b) Statement is correct and reason is false

c) Both statement and reason are false

d) Statement is false, reason is correct

8. Find out the correct statements from the following based on the relationship between hypothalamus and the pitutary gland.

Statement 1 : The hormones produced by the hypothalamus are stored to the posterior lobe of the piturary and released as required.

Statement 2 : The hormone production in the anterior lobe of the pitutary is controled by tropic hormone produced by hypothalamus.

Statement 3 : Oxytocin and vasoprassin are the tropic hormone produced by hypothalamus.

Statement 4 : The function of thyroid gland is controlled by the tropic hormone produced by the pitutary.

a) All statements are correct

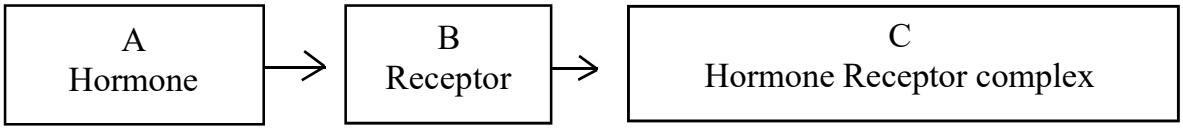
b) Statement 1, 2 and 4 are correct

c) Statement 2, 3 and 4 are correct

d) Statement 1, 3 and 4 are correct

2 Mark Questions

1. A diagram related to the arrival of hormones at target cells is given. Analyse it and write the answers to the questions.



a) Explain the function of 'A' in account to the formation of 'C'?

b) If 'A' is the releasing hormone, where 'B' is found?

c) How do the production and transportation of 'A' differ in plants and animals?

d) What is the cell on which 'A' acts called?

2. A description of plant hormone is given in the box. Analyse it and answer the questions below.

> Stimulate lateral bud growth.
>
> Delays leaf aging and maintain greeness

a) Which plant hormone is mentioned in the box?

b) Write another function of this plant hormone?

3. Find and write the correct statements related to the regulation of calcium level in the blood.

i) Parathormone helps to increase calcium level

ii) Calcitonin accelerates the storage of calcium in the bones

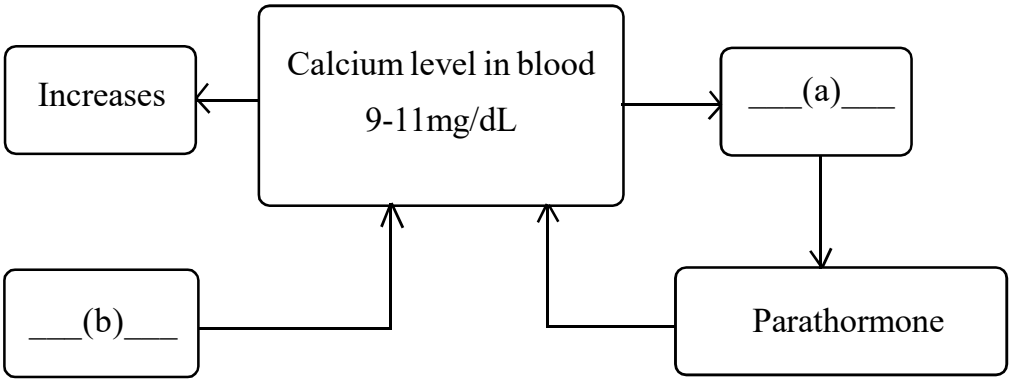
iii) When the calcium level in the blood increases the parathormone glands stimulated

iv) Parathormone increases the absorption of calcium through the intestine..

v) Calcitonin increases the reabsorption of calcium in the kidneys

vi) Parathormone is produced by the thyroid gland

4. The hormonal activities related to the regulation of the calcium level in the blood are given below. Analyse and answer the questions.



a) Name the hormones which indicate a & b?

b) Redraw this illustration based on glucose.

5. Analyse the statement and reason and find the correct one.

Statement : When the glucose level in the blood decreases, the alpha cells of the pancreas produce insulin.

Reason : The hormone insulin accelerates the entry of glucose in to the cell.

a) Both statemet and reason are incorrect

b) Statement is correct, reason is incorrect

c) Statement is incorrect, reason is correct

d) Both statement and reason are correct

6. Find the incorrect statement regarding the hormone Somatotropin.

i) Helps in body growth

ii) Reabsorbs water from the kidneys

iii) Stimulates the thyroid gland

iv) Plays an important role in immunity

v) Stimulate the function of the ovaries and testes

vi) Does not stimulate the adrenal cortes

3 Mark Questions

1. 'All hormones do not act on all cells'

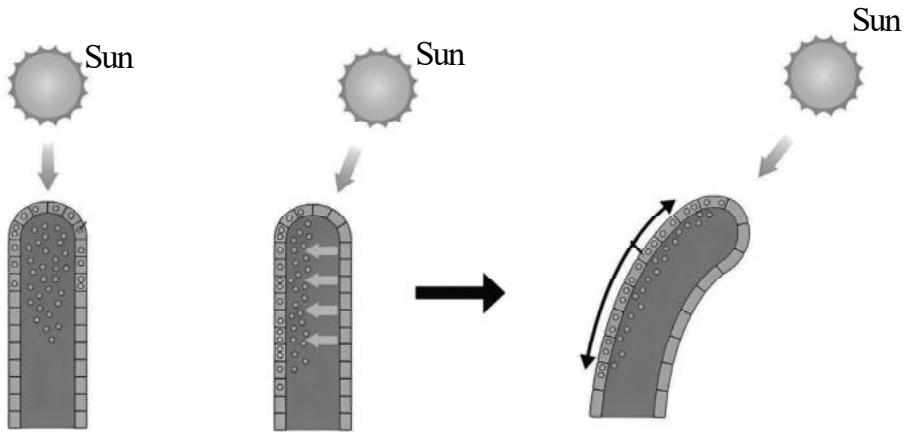
   Analyse the statement and write the answers to the questions.

   a) Write the reason for the statement (1)

   b) Write the differences observed in hormone transportation in animals and plants. (2)

2. Prepare a short not on Hormone. Target cell and Receptor complex.

3. Observe the illustration and answer the questions.



   a) Which hormone helps in the elongation of cells?

   b) Write the changes that occur in the plant stem when sunlight is received only from one direction.

   c) What is the reason for the plant stem growing towards the light?

4. Analyse the table and answer the questions.

Hormone	Function
Auxins	Increases stem length
Gibberllins	(a) _____
(b) _____	Maintain seed dormancy

a) Write the answer for a and b?

b) Find the difference in the action of Gibberellins on seeds and fruits?

c) Which hormone helps plants cope with adverse conditions? How does they act?

5. Analyse the statements and answer the questions.

   i) Follicular cells produce calcitonin

   ii) Para follicular cells produce Thyroxine

   iii) Follicular cells produce thyroxine

   iv) Para follicular cells produce calcitonin

   1. Find the correct statements?

   2. Write the functions of thyroxine produced by thyroid gland?

   3. Name the condition caused by less production of Thyroxine?

      Write any two of its symptoms?

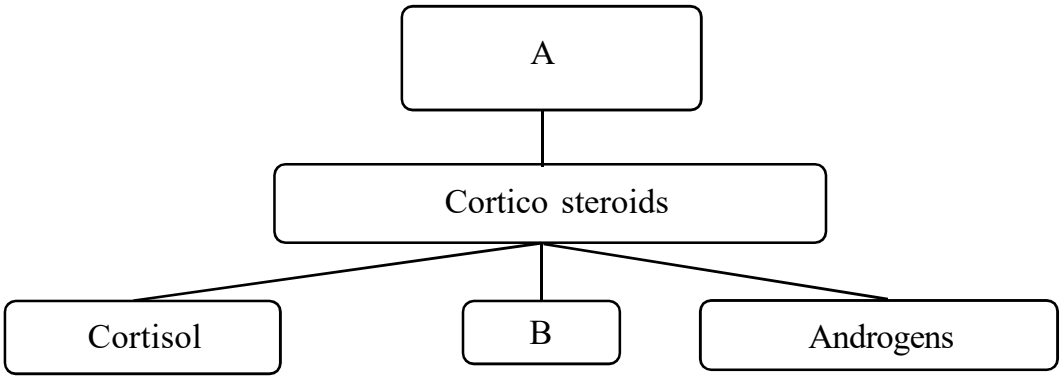
6. The pancreas is a gland of equal importance in both the Digestive system and endocrine system.

   i) Name the group of cells which produce hormones in pancreas. What are the two types of cells there?

   ii) Write the normal level of glucose in blood?

   iii) What is the disease condition where the blood glucose level rises from the average (126mg/dL). What are the symptoms of this disease.

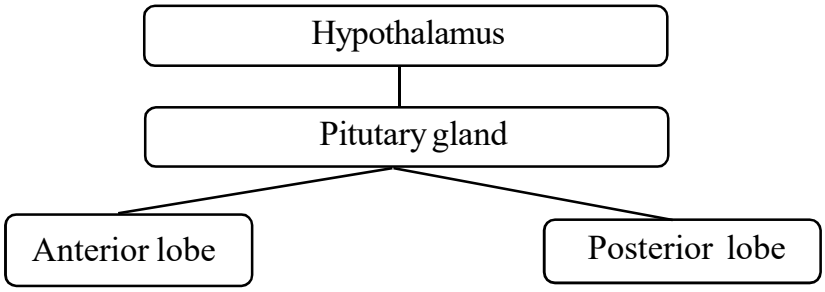
7. Observe the illustration and write down the answers



a) Find A and B?

b) What is the inner part of Adrenal gland? What are the hormones produced?

c) Why any function of control?



a) What are the part that help to transport the hormones produced by the hypothalamus to the pitutary gland?

b) Compare the function of releasing hormones and inhibitory hormone?

c) What are the hormones stored by the posterior lobe of the pitutary gland?

4 Mark Questions

1. Analyse the table and find the answers to the questions below.

Disease	Cause	Symptoms
a) Dwarfism	Increased production of Somatotropin during growth period.	Excessive growth of body parts like palms, legs and jaw.
b) Gigantism	Increased production of Somatotropin during the growth period	Excessive body growth
c) Acromegaly	Decreased production of Somatotropin during the growth period	Stunted body growth

1) Select and write the correct pair? (1)

2) Which is the endocrine gland that produce the growth hormone Somatotropin? (1)

3) Re-arrange and write the incorrect pairs? (2)

2. 'Ants travel in a line is due to some chemical secreted by those organism'

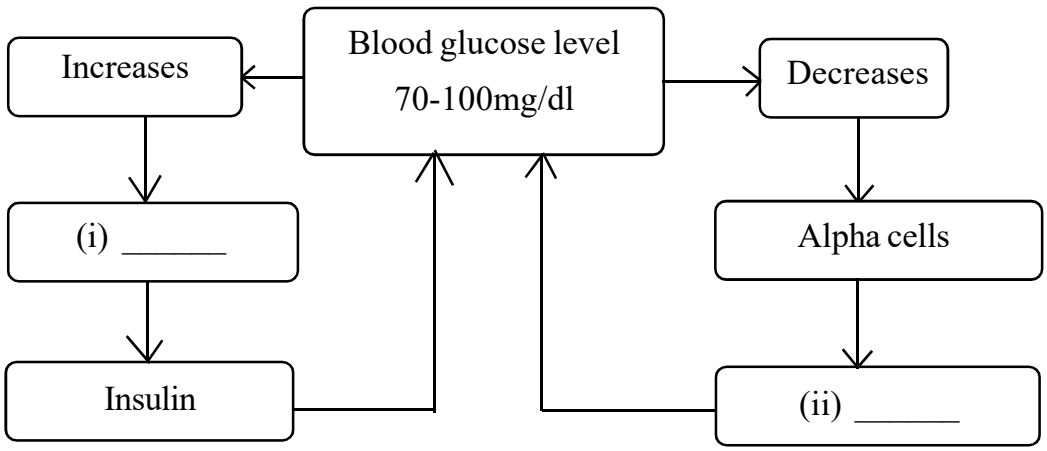
1) What is the name of these chemicals? (1)

2) Write any two other function of these chemicals? (1)

3) Write two type of such chemical? (1)

4) How do farmers utilize this chemical? (1)

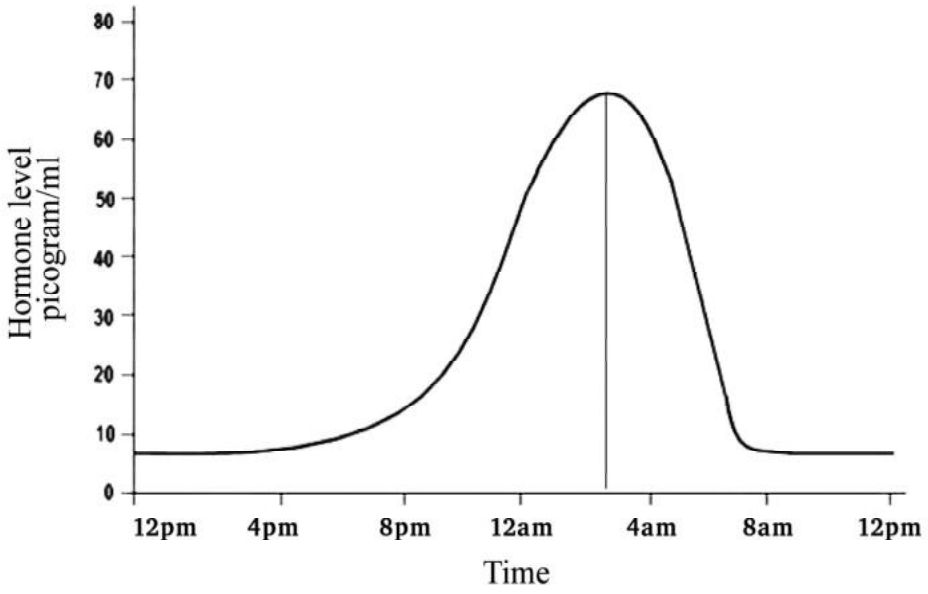
3. Answer the following.



a) Identify and write i, ii? (2)

b) Redraw the illustration showing calcium instead of glucose?
(Blood calcium level 9-11mg/dl) (2)

4. Observe the given graph and write the answers to the questions given below.



a) Which hormone indicated in the graph, influences sleep and wakefulness? (1)

b) At what time is the production of this hormone increased? (1)

c) Which gland produces this hormone? Where is it located? (1)

d) What is the necesity of avoiding mobile phone usage before sleeping? (1)

5. Fluctuations in Thyroxine production and malfunction of the gland lead to diseases conditions. Analyse the table below and write the answers to the questions.

Condition	Causes	Symptoms
Hyper thyroidism	(a) _____	Increase in the rate of metabolism, weight loss, heart beat increases, excessive sweating anxiety
(b) _____	Reduced Secretation of thyroxine	Metabolism decreases, body weight increases, heart rate slows down, fatigue, intolerance to cold
Goitre	(c) _____	(d) _____

1) Write a, b, c & d?

2) What are the two type of cells present in Thyroid gland?

3) Which cell produces thyroxine?

6. ADH heps to maintain the water balance in the body. Classify the physiological changes experienced during summer and winter by giving appropriate headings.

- Production of ADH increases
- Production of ADH decreases
- Reabsorption of water by kidneys increases
- Reabsorption of water by kidneys decreases
- Urine production decreases
- Urine production increases

Chapter 4

CHEMORECEPTION IN ORGANISMS

ANSWERS

1 Mark

1. c) ii), iii) are correct
2. a) ii) - v), iii) - iv)
3. b) Target cells
4. a) Both statements are correct
5. a) Statement is correct, reason is false
6. c) Both statement and reason are correct
7. a) Both statement and reason are correct
8. b) Statement 1, 2 and 4 are correct

2 Mark

1. a) Hormones combine with their receptors which is present in their receptors which is present in the cell membrance and with in the cell, to form hormone receptor complex. This complex controls activities inside the cell.
   b) Anterior lobe of pitutary
   c) Plant hormones are produced at the top of the shoot or root, in seeds or sprouting leaves. They reach the target cell via cell to cell tramport through xylem and phloem. In animals, the hormones produced by endocrine land reach the target cell through blood.
   d) Target cells

2. a) Cytokinins
   b) Promote the transport of nutrients to growing region

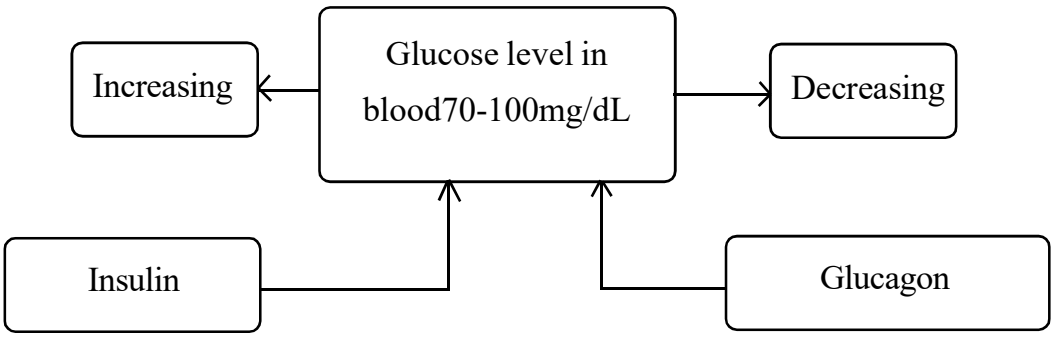
3. i) Parathormone helps to increase calcium level
   ii) Calcitonin accelerate the storage of calcium in the bones

iv) Parathormone increases the absorption of calcium through the intestine..

v) Calcitonin increases the reabsorption of calcium in the kidneys

4. a) Decreasing b) Calcitonin

b)



5. c) Statement is incorrect, reason is correct

6. ii) Reabsorbs water from the kidneys

iii) Stimulates the thyroid gland

iv) Plays an important role in immunity

v) Stimulate the function of the ovaries and testes

3 Mark

1. a) The target cell of each hormone has receptors suitable for that hormone

b) In animals, the hormones produced by endocrine glands reach the target cell through blood. In plants the hormones are produced at the tip of the shoot or root, in seeds or in sprouting leaves. They reach the target cell via cell to cell transport or through xylem and phloem.

2. Hormones are the chemical messengers that control vital activities in plants and animals. Their deficiency disrupt vital activities. Hormone always acts on its target cells. Each hormone has specific receptors in its target cell.

Hormones and receptors together form a hormone receptor complex. This combination controls the activities with in the cell.

3. a) Auxins

b) When light falls on the stem from only one direction, auxin moves to the opposite side of the stem. There auxin combines with the hormone receptors in the cell to form a hormone receptor complex and as a result, changes occur in cellular activities.

c) The cells on the shaded side elongate and the part where cell elongation occurs bends towards the opposite side.

4. a) a) Break seed dormancy b) Abscisic acid

b) In seeds, gibberellins break seed dormancy and also influence the growth of fruits and seeds.

c) Abscicic acid

d) Inhibits the growth of plant part parts and enables them to tide over unfavourable conditions. It also promotes abscission of mature leaves and fruits.

5. 1. iii, iv

2. * Controls metabolism and body temperature
   * Development of brain in newborn babies and children
   * Helps to maintain the health of the heart, skin, reproductive system

3. Hypothyroidism

   Symptoms : Increases the rate of metabolis weight loss, heart beat increases, excessive sweating, anxiety.

6. i) Islets of Langerhans, Alpha cells, Beta cells

ii) 70 - 100mg/dL

iii) Diabets Mellitus

Symptoms

* Excessive thurst
* Hunger
* Frequent urination
* Weight loss
* Fatigue

7. a) A - Cortex

   B - Aldosterone

   b) Medulla

   Epinephrine

   Nor epinephrine

   c) * Maintain glucose homeostasis

   * Supresses the immune responses

   * Resists inflammation and allergy of the body

8. a) Neurosecretary cells and portal vein

   b) Releasing hormone reaches the anterior lobe of the pitutary and stimulate the production of tropic hormone. Inhibitory hormone inhibit their production.

   c) Oxytocin, Vasopressin

4 Mark

1. 1) b

   2) Anterior lobe of pitutary

   3)

Disease	Cause	Symptoms
Dwarfism	The production of somatotropin decreases during growth phase	Stunted growth Excessive growth of body
Acromegaly	The production of somatotropin increases after growth phase	parts such as palm, foot & jaws

2. 1) Pheromones

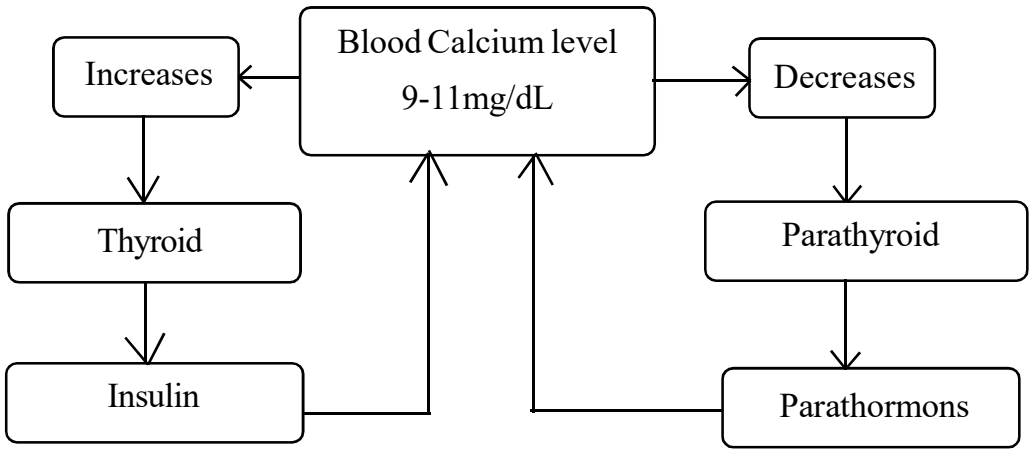
   2) To determine territory, to attract mates

   3) Territorial pheromone, Aggregation pheromone, Alarm pheromone, Trail pheromone, sex pheromone

   4) Farmers using pheromons trups for pest control

3. a) i) Beta cells ii) Glucagone

b)



4. a) Melatonin
   b) Between 12 am and 4 am
   c) Pineal gland, located near thalamus
   d) Mobile phone should be completly avoided before sleeping, then only melatonin is produced in the required quantity for the brain.

5. 1 a) Production of thyroxine increases
   b) Hypothyroidism
   c) Malfunctioning of thyroid gland, deficiency of iodin
   d) Swelling in the neck region, difficulty in swallowing
   2) Follicular cells and Parafollicular cells
   3) Follicular cells

6. *Summer Season*
   * Production of ADH increases
   * Kidney's increases the reabsorption of water
   * Production of urine decreases

   Winter
   * Production of ADH decreases
   * Kidneys decreases the reasbsorption of water
   * production of urine increases

Chapter 5

IMMUNITY AND HEALTHCARE

1 Mark Questions

1. Statement A : Antigens are anything including pathogens which enter the body.

   Statement B : Immunity is the ability of the body to prevent the entry of pathogens.

   Based on the above statements choose the correct answer.

   a) A & B are correct

   b) A & B are incorrect

   c) A is right B is incorrect

   d) A is wrong B is right

2. Statement A : Environmental factors influences health

   Statement B : Polluted air and contaminated water adversly affect health.

   Based on the above statements choose the correct answer.

   a) A & B are correct

   b) A & B are incorrect

   c) A is right B is incorrect

   d) A is wrong B is right

3. Assertion (A) : There is a special system in our body to prevent diseases.

   Reason (R) : Antibody help us to identify pathogens which enter in our body.

   Based on the above statements, choose the correct answer from the following option.

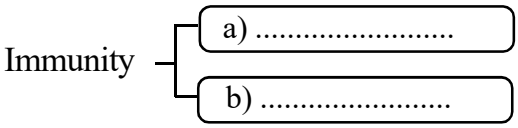
   a) A & R are correct

   b) A & R are correct

   c) A is incorrect, R is correct

   d) A is right R is incorrect

4. Complete the table



5. Choose the correct statements related to Innate immunity from the following statements.
   a) Innate immunity is developed after birth
   b) The natural immune mechanism that exists in the body by birth is caled Innate immunity.
   c) Innate immunity is also known as primary level immunity
   d) Lymphocytes plays major role in acquired immunity

6. Find out the correct order of reactions in phagocytosis from the following options.
   a) Expulsion of waste materials
   b) Engulfment of pathogens in phagosome
   c) Enzymes destroy pathogens
   d) Phagocyte approaches pathogens
   e) Phagolysosome is formed

   i) d, c, e, b, a ii) d, b, e, c, a
   iii) b, c, e, a, d iv) b, e, c, a, d

7. Find out those related to B - lymphocytes from the following and choose the correct option.
   a) Production of antibodies
   b) Destruction of cancer cells
   c) Maturation in the thymus gland
   d) Formation in the bone marrow

   i) a, b and d are correct
   ii) a, c and d are correct
   iii) b and c are correct
   iv) a and d are correct

8. Which of the following is the correct pair of Phagocytes?

a) Eosinophil, Neutrophil

b) Basophil, Neutrophil

c) Neutrophil, Monocyte

d) Monocyte, Lymphocyte

9. Which of the following diseases are prevented by MMR vaccine?

a) Polio b) Tuberculosis c) Measles

d) Rubella e) Covid f) Mumps

i) a, b, c ii) c, d, f

iii) b, c, d iv) d, e, f

10. Analyse the statements and choose the correct answer from the following.

Statement : Immunisation is an artificial method of preparing the immune cells in advance expecting the attack of pathogens.

Reason : The natural immune mechanism that exists in the body by birth is called Innate Immunity.

a) Statement and Reason are correct

b) Statement is incorrect, Reason is correct

c) Statement and Reason are incorrect

d) Statement is correct, Reason is incorrect

11. Which of the following options are the examples of Contageous diseases?

a) Diabetes b) Leptosprosis c) Nipah

d) Stroke e) Tuberculosis f) AIDS

i) a, b, c, d ii) b, c, d, e

iii) c, d, e, f iv) b, c, e, f

12. Choose the examples of bacterial diseases from the following options.

a) Tuberculosis	b) Nipah	c) AIDS
d) Leptospirosis	e) Malaria	

i) a and d ii) a, b and c
iii) d and e iv) a, d and e

13. Identify the disease and disease causing organism from the symptoms given below
* White patches in the mouth
* Itching, Thick discoloured nail

14. Complete the table

Disease	Causative Organism	Mode of Transmission
X	Y	Female Anopheles Mosquitto

15. Analyse the given symptoms and find out the plant disease and what type of micro organisms cause the diseases?
a) Round yellow spots on the leaves
b) Spots on the rice grain

16. If a person is having blood group O negative, name the antigen and the Rh factor present in this blood.

17. Analyse the given statements given below and correct the underlined part, if mistakes are there.
a) Antipyretic drugs are used for the treatment of worms
b) The drugs used for the treatment of fever are Antihelminthic medicines

18. Choose the correct pair from the following

a) Cardiology	i) Heart
b) E N T	ii) Ear, Nose, Throat
c) Urology	iii) Children
d) Gynaecology	iv) Kidneys

19. Analyse the given symptoms and find out the disease and name the causative organism of the disease.
    a) Intemittent fever, chills and shivering
    b) Headache, Fatigue and body pain

2 Mark Questions

1. Complete the table

Blood Group	Antigen	Antibody
A	A	(i) ____
B	B	(ii) ____
AB	(iii) ___	Nil
O	(iv) ___	a, b

2. Analyse the given table and find out the correct statements and reasons.

Statment	Reason
i) Antibiotics are medicines that are either derived from micro organisms or produced synthetically and are used against bacteria	When the control mechanisms of the cell division process fail, normal cell turns into cancer cells.
ii) The type of antigen present in the blood determines the blood group	Blood groups are determined by the presense of A and B antegen present on the surface of RBCs.
iii) Cancer is a condition in which abnormal and uncontrolled cell division leads to multiplication of cells	They are the most effective medicines available to treat bacterial diseases.
iv) In Sickle cell anaemia RBCs become sickle shaped	Defect of the gene that controls the production of Haemoglobin

3. Complete the table

Disease	Causative Organism	Affected Organisms
Ranikhet disease	Virus	(i)_____
Foot mouth disease	(ii)_____	Cattles
Anthrax	(iii)_____	Cattles
Mastitis	(iv)_____	Cattles

4. Analyse the table showing certain pathogens in an area and the number of people affected by them and answer the questions.

Pathogen	No. of injected people
Nipah virus	10
Micro bacterium tuberculosis	40
Leptospira	08
Naegleria Fowleri	02

i) Which disease affected more people in the region? Which group of micro organisms do the pathogen belongs to?

ii) Which disease is affected the residents who bathed in stagnant fresh water? Which group of micro organisms do the pathogen belongs to?

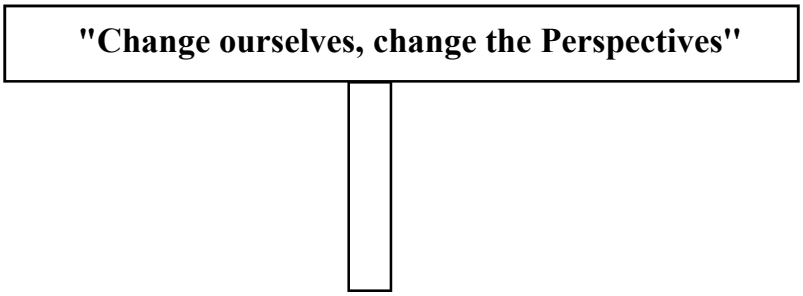
5. Write the functions of following terms related to the immunity in plants.

i) Cell Wall ii) Enzymes

iii) Callose iv) Antimicrobial Substances

6. "Although antibiotics are effective medicines, their irrational use is harmful" Justify the statement.

7. The following text is from a placard prepared in connection with World AIDS day.



i) Write any two methods of transmission of this disease?

ii) Approach toward those affected by the disease .

8.



A



B

a) Identify A and B?

b) Write the symptoms of A and B?

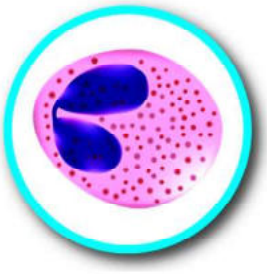
9. Complete the table

Part of the body	Immune Mechanism
Mouth	a)
Stomach	b)
Trachea	c)
Ear	d)

10. Identify A, B, C and D?



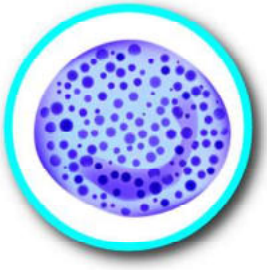
A



B

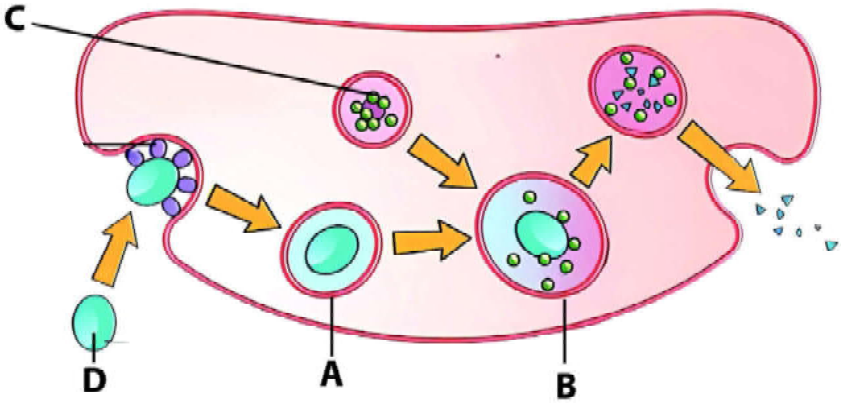


C



D

11. Identify A, B, C and D?



12. Correct the statements, if any mistakes in the underlined parts.

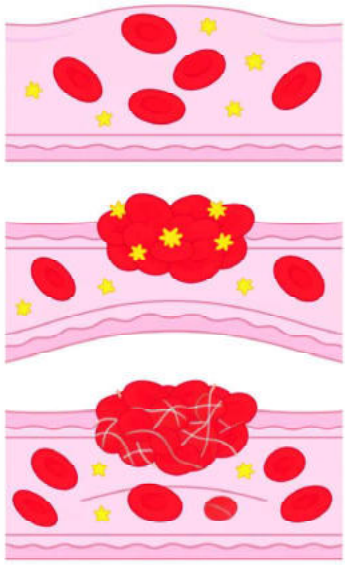
a) Fever is the condition that occurs when the body temperature rises above the normal range.

b) When infection or inflamation occurs, white blood cells release chemical substance called Enzymes.

c) Pyrogens trigger the brain to raise the body temperature

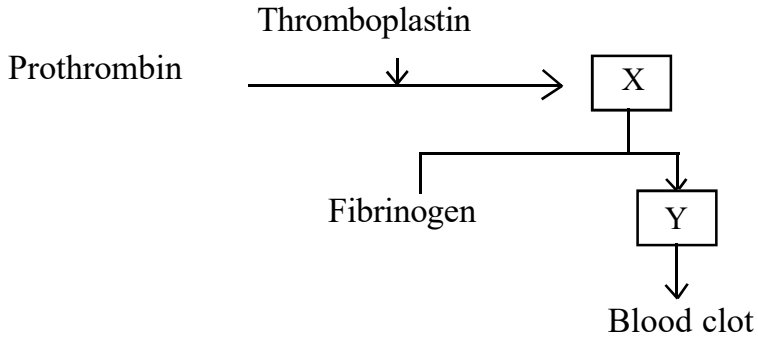
d) High temperature helps lymphocytes to destroy pathogens

13. Analyse the illustration and answer the following questions.



a) Identify the illustration. Name the blood cell which produces thromboplastin?

b) Complete the flow chart.



14. Complete the table

Cells in specific Immunity	T-lymphocytes	B-lymphocytes
Formation	(a) _____	Bone marrow
Maturation	(b) _____	(c) _____
Function	(d) _____	Produces antibodies and destroy pathogens

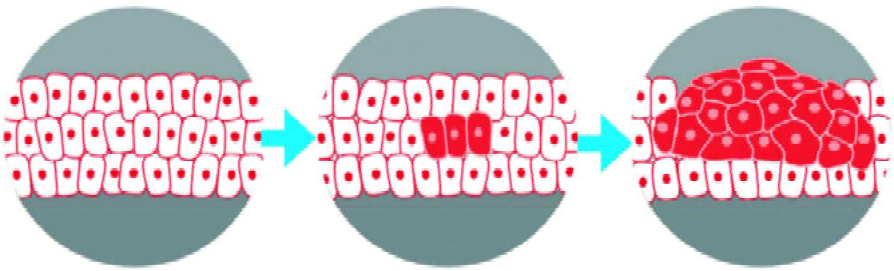
15. Complete the table

Vaccine	Preventable diseases
MMR	Measles, ___(a)___, Rubella
OPV	b) _____
HPV Vaccine	c) _____
Covid 19 Vaccine	d) _____

16. Complete the Table

Type	Diseases
Lifestyle diseases	i) _____
ii) _____	Silicosis
Autoimmune Disorders	iii) _____
Nutritious Disorders	iv) _____

17. Illustration A.

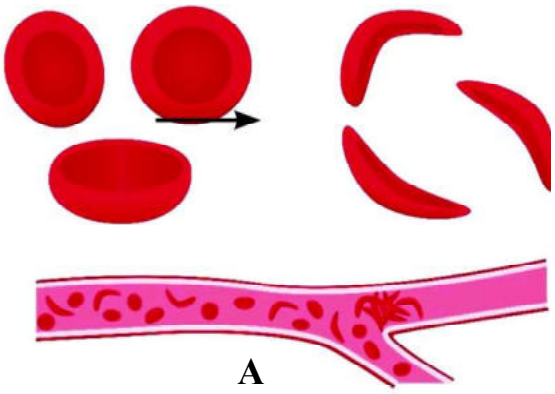


Identify the illustration A. Write three causes of A?

18. Arrange the table correctly

Equipment	Use
Stethescope	Oxygen level
Digital BP Apparatus	Internal Organs
Pulse Oxymeter	Blood Pressure
Endoscope	Heart beat

19. Analyse the illustration and answer the following questions.



A

a) Identify the illustration A and which type of blood cells undergo change in its shape?

b) What is the cause of A?

20.



A

a) Identify the illustration A. Name the causative organism of A?

b) Write the mode of transmission of A?

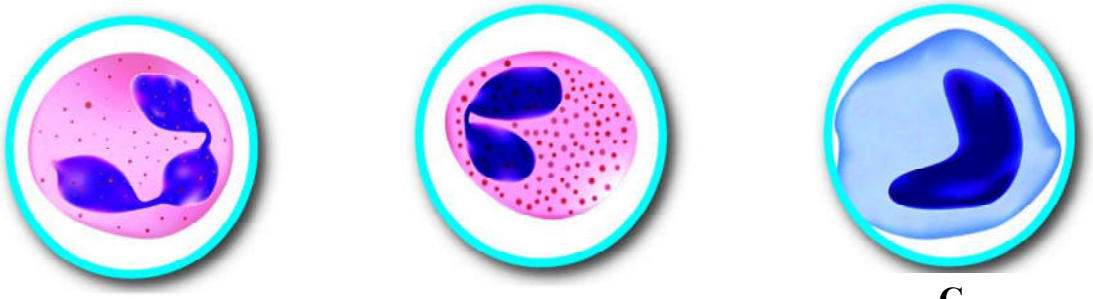
3 Mark Questions

1. 'Pathogens around us cause different diseases'.
   Analyse the given statement and find out what are the ways the pathogens enter our body?

2. Complete the table

Parts of the Body	Immune Mechanism
Nose	Mucus
Eye	i) .....
ii)	HCl
Tachea	iii) .....
Skin	iv).....
Eay	v) .....
Mouth	vi) .....

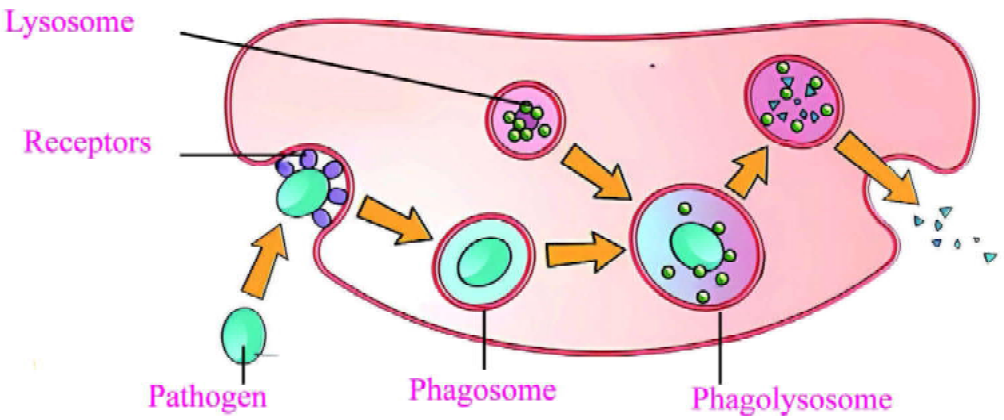
3.



A **B** **C**

Analyse the illustration and identify A, B, C and write the functions of each.

4.

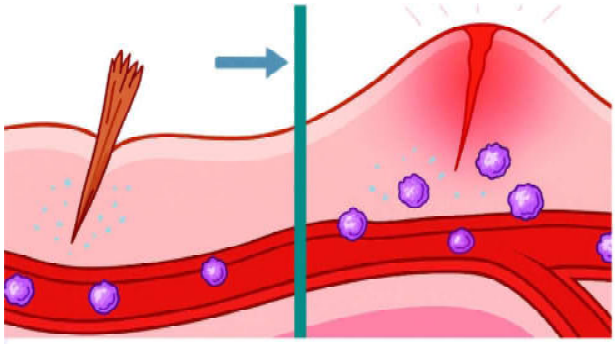


a) Identify the illustration?

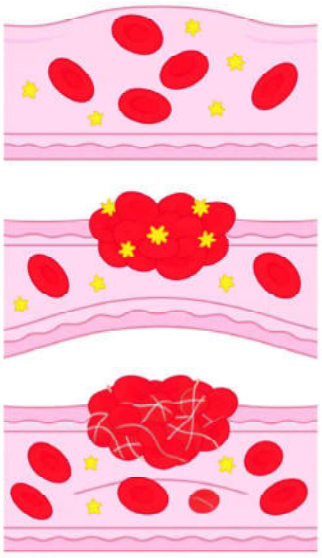
b) Write the importance of Lysosome in the illustration?

c) Name the two blood cells participating in the process shown in the illustration.

5.



A



B

a) Identify A and B in the illustration?

b) Name the blood cells which produce Histamine and Thrombiplastin in the illustration A and B.

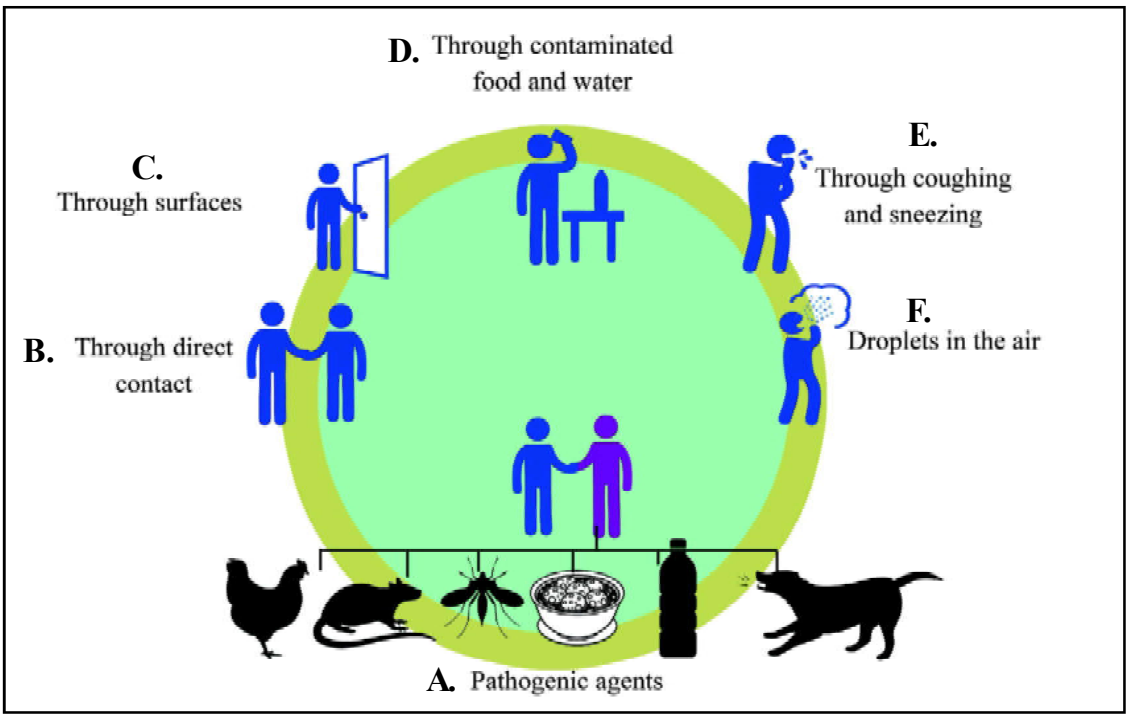
c) What is the importance of the formation of Fibrin threads in the illustration B?

6. Complete the table

Vaccine	Components of vaccine	Preventable diseases
MMR	a)	Measles, Mumps, b) .....
OPV	Contains dead pathogens	c) .....
HPV Vaccine	d)	e) .....
Covid-19 Vaccine	Contains parts of the genetic material of the pathogen	f) .....

7. Explain the action of vaccines in our body?

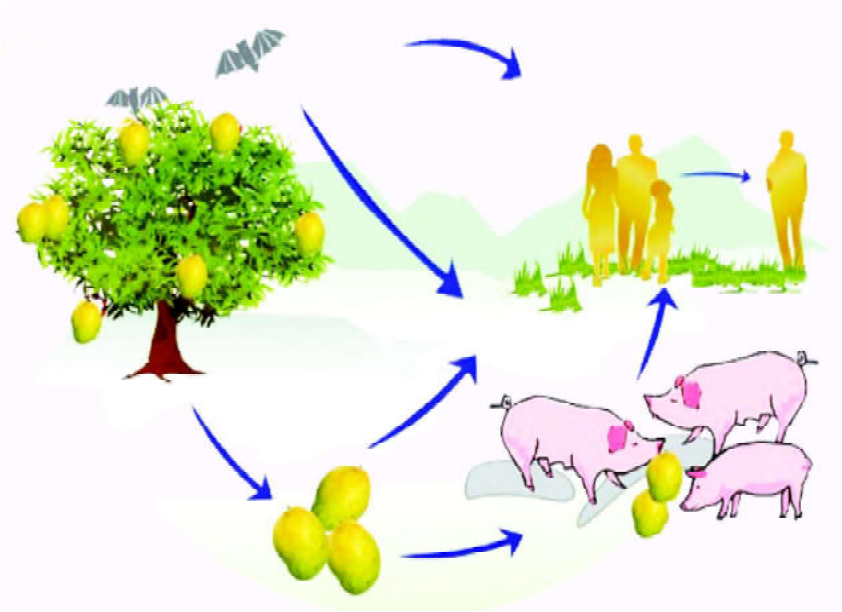
8. Analyse the illustration about modes of disease transmission. Identify any one disease transmitted through the each modes of disease transmission.



9. Symptoms of a disease is given below. Analyse it and write the name of the disease, disease causing organism, mode of transmission, part of the body affected and preventive measures.

Persistent cough for more than two weeks, fever, loss of appetite, weight and fatigue...

10.



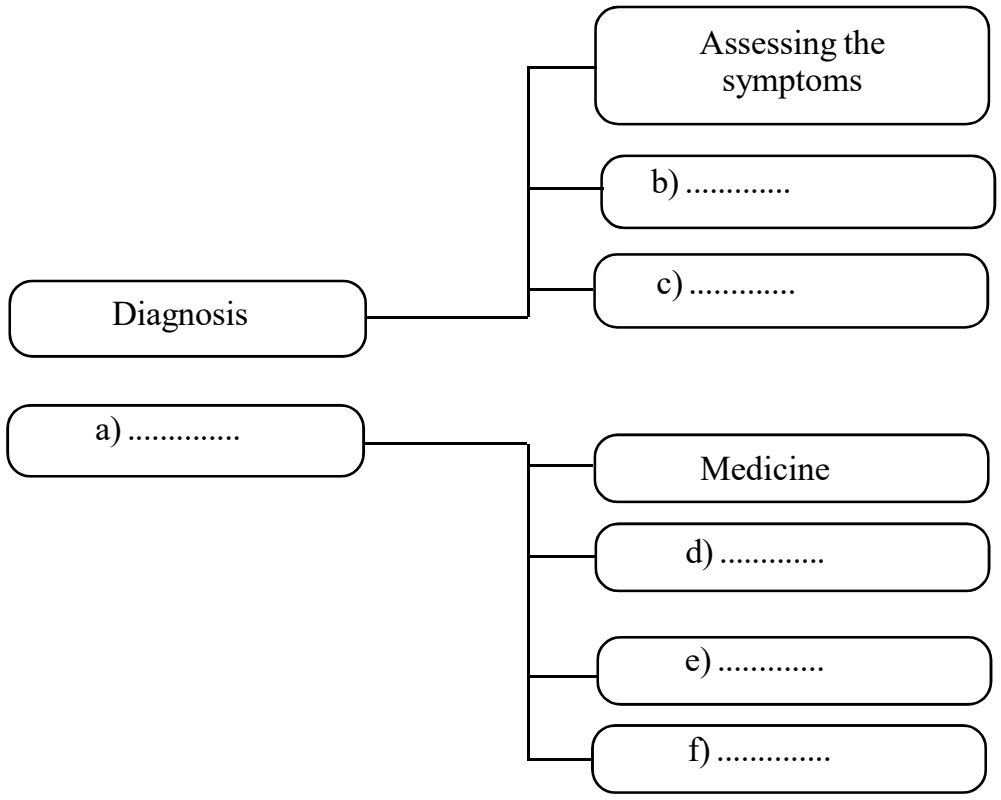
a) Identify the disease illustrated above and find out the causative organism.

b) Write the symptoms of this disease.

c) Write the mode of transmission and preventive measures of the disease.

11. What are the treatment methods of cancer?

12. Complete the flow chart

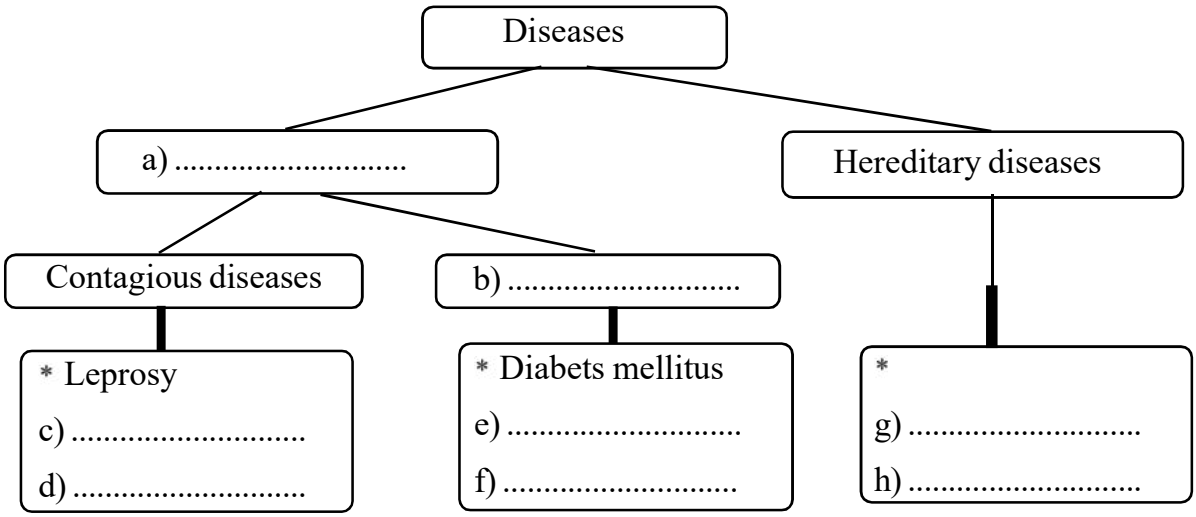


13. Complete the table

Blood group	Antigen	Antibody
A	A	(i)
B	(ii)	a
AB	(iii)	(iv)
O	(v)	(vi)

4 Mark Questions

1. Complete the flow chart.



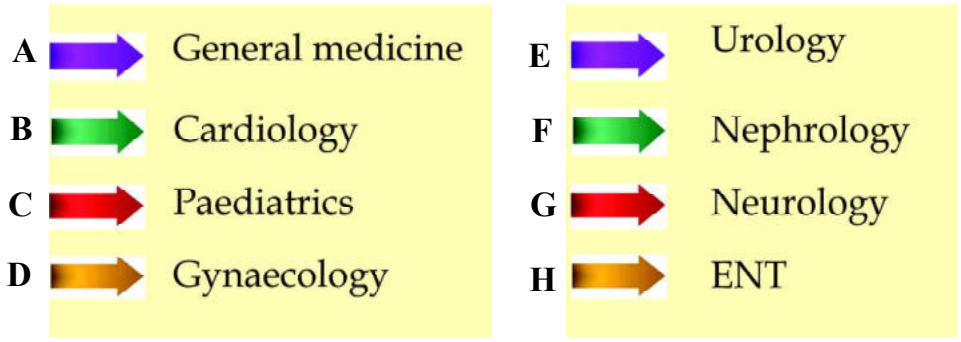
2. Complete the table

Disease	Pathogen
Tuberculosis	i) .....
ii)	Nipah Virus
Rat Fever	iii) .....
AIDS	iv) .....
v)	Filarial worm
Malaria	vi) .....
Amoebic Encephalitis	vii) .....
viii)	Fungus

3. Analyse the table and Find out a, b, c, d, e, f, g & h.

Equipment	Use
Stethescope	a) .....
Sphygmomanometer	b) .....
Digital BP apparatus	c) .....
Pulse oximeter	d) .....
Thermometer	e) .....
Endoscope	f) .....
US scanner, CT scanner	g) .....
MRI Scanner	h) .....

4. Analyse the board displayed in the hospital. Find out what each one is related to.



5. 'Use antibiotics only as prescribed by a doctor'. Justify the statement.

6. Find out the disease and causative organism of diseases from the given symptoms.
   a) Intermittent fever, chills and shivering, headache, vomiting, muscular pain, body pain and fatigue
   b) White patches in the mouth itching, thickened nails
   c) Immunity decreases, pneumonia, cancer, tuberculosis
   d) Sudden onset of fever, headache, muscle pain, redness of the eye, vomiting

7. Prepare a seminar topic based on the topic immunity in plants.

8. 'It is necessary to consider the blood group in blood transfusion'.
   Analyse the statement and answer the following questions.
   a) What is blood transfusion?
   b) Write any two guideline for donor in blood transfusion.
   c) What is the basis of blood group?
   d) If a person is having the blood group 'O' negative, write the antibodies and the Rh factor in the person?

9. Find out the disease and causative organism from the given symptoms.
   a) The tender leaves turn into yellow and the crown of the tree troops and falls.
   b) Yellow and white coloured patches on leaves, growth retards and the size of tuber decreases.
   c) Round yellow spots in leaves and grains.
   d) Leaves become short, narrow and stand upright, growth becomes retarded

10. 'Innate immunity includes various mechanisms to prevent antigens from entering the body'.

Analyse the statement given above and answer the following questions.

a) Name the WBCs which take part in innate immunity?

b) List out four mechanisms in Primary Level Immunity.

Chapter 5

IMMUNITY AND HEALTHCARE

ANSWERS

1 Mark

1. a) A & B are correct
2. a) A & B are correct
3. a) A & R are correct, R is the correct explanation of A
4. a) Innate Immunity
   b) Acquired Immunity
5. b) The natural immune mechanism that exists in the body by birth is caled Innate immunity.
   c) Innate immunity is also known as primary level immunity
6. ii) d, b, e, c, a
7. iv) a and d are correct
8. c) Neutrophil, Monocyte
9. ii) c, d and f
10. d) Statement is correct, Reason is wrong
11. iv) b, c, e and f
12. i) a and d
13. Candidiasis, Fungus
14. X - Malaria
    Y - Plasmodium (Protozoa)
15. Blight disease of Paddy, Bacteria
16. A B Antigens are absent, Rh antigen absent

17. a) Anithelminthic

    b) Anti pyretic

18. a, b

19. Malaria, Plasmodium

2 Mark

1. i) b ii) a
   iii) A, B iv) Nil

2. ii
   iv

3. i) Chicken / Duck ii) Virus iii) Bacteria iv) Bacteria

4. i) Tuberculosis, Bacteria
   ii) Amoebic Encephalitis, Protozoa

5. i) Prevent the entry of pathogens
   ii) Breakdown the cellwalls of pathogens
   iii) Prevents spreading of pathogens
   iv) Kill pathogens or inhibit their growth

6. Yes. Irrational use of antibiotics are harmful and causes several side effects such as Allergy, damage to the liver and kidney, problems in digestive system, destruction of beneficial bacteria in the intestine.

7. i) HIV infection can be transmitted in various ways
   * Usage of unsterilized needles, syringe
   * Sexual contact with injected blood
   * From HIV infected mother to foctus

   ii) HIV injection does not transmit during speaking, sharing food, room and toilets, ...
   So the society should give maximum support to HIV injected person.

8. a) A - Ring Worm B - Candidiasis
   b) A - Circular reddish rashes
      B - White patches in the mouth, itching

9. a) Saliva
   b) HCl
   c) Cilia
   d) Ear wax

10. A - Neutrophil B - Eosinophil
    C - Monocyte D - Basophil

11. A - Phagosome B - Phagolysosome
    C - Lysosome D - Pathogen

12. a) Fever
    b) Pyrogens
    c) Pyrogens
    d) Phagocytes

13. a) Blood clotting, Platelets
    b) X - Thrombin
       Y - Fibrin

14. a) Bone Marrow
    b) Thymus gland
    c) Bone Marrow
    d) Destroy Cancer cell

15. a) Mumps
    b) Polio
    c) Cervical Cancer
    d) Covid

16. i) Diabets/High BP
    ii) Occupational diseases
    iii) Rheumatoid Arthritis
    iv) Anaemia

17. a) Cancer cell growth

Genetic changes, Environmental factors, Viral infections, Lifestyle

18.

Equipment	Use
Stethescope	Heart beat
Digital BP Apparatus	Blood Pressure
Pulse Oxymeter	Oxygen level
Endoscope	Internal Organs

19. a) Sickle Cell Anaemia, RBC

b) Defect of the gene that controls the production of haemoglobin

20. a) Filariasis, Filarial worm

b) Female anopheless mosquitoes

3 Mark Answers

1.
- Through direct contact
- Through droplets in air
- Through contaminated food and water
- Through cough and sneezing
- Through unsterilised syringes, blood transfusion

2. i) Lysozyme in tears
ii) Stomach
iii) Cilia
iv) Sweat/Sebum
v) Ear wax
vi) Lysozyme in saliva

3. A - Neutrophil - Engulfs and destroy bacteria, produces chemicals against pathogens

B - Eosinophil - Neutralizes chemicals released when there is an allergy, destroying parasites.

C - Monocytes - Engulfs and destroys foreign substances, bacteria and dead cells.

4. a) Phagocytosis
   b) Enzymes in the Lysosome destroy pathogens
   c) Neutrophil, Monocyte

5. a) A - Inflammatory responses
      B – Blood clotting
   b) Basophils, Platelets
   c) Fibrin forms a network of fibers that trap blood cells, forming a blood clot.

6. a) Contains neutralized pathogens
   b) Rubella
   c) Polio
   d) Contains parts of pathogens
   e) Cervical cancer
   f) Corona

7. * Vaccines acts as antiens in our body
   * Stimulate our immune system
   * It produce antibodies
   * Antibodies remain in our body and provides long term protection against diseases.
   * Vaccines provide protection from a disease for many years or even for a lifetime.

8. A - Malaria / Rat Fever
   B - Fever / Influenza
   C - Ring worm
   D - Cholera / Diarrhea / Typhoid
   E - TB / Corona / Influenza
   F - TB / Fever / Influenza

9. TB, Mycobacterium tuberculosis

   By breathing tiny droplets released into the air when the patient is coughing, sneezing
   Lungs, Kidneys, Bones, Brain..

   BCG vaccine

10. a) Nipah, Nipah virus
    b) Fever, headache, muscular pain, cough, vomitting, disorientation, epilepsy and loss of consciousness
    c) Bats, Pigs. There are currently no antiviral medicine or vaccines.

11. Surgery, Chemotherapy, Radiation therapy, Immunotherapy, Targeted therapy, Hormone Therapy

12. a) Treatment
    b) Laboratory Test
    c) Body check up
    d) Surgery
    e) Organ transplantation
    f) Radiation therapy

13. (i) b
    (ii) B
    (iii) A & B
    (iv) Nil
    (v) Nil
    (vi) a, b

4 Mark Answers

1. a) Acquired diseases
    b) Non contagious diseases
    c) & d) Ring worm, Rat Fever, Tuberculosis, Candidiasis, AIDS, Nipah (any two)
    e) & f) Stroke, Heart attack, blood pressure, anaemia, silicosis, rheumatoid arthritis (any two)
    g) & h) Sickle cell anaemia, Haemophilia

2. i) Myco bacterium tuberculosis
    ii) Nipah
    iii) Leptospira
    iv) HIV

v) Filariasis
vi) Plasmodium
vii) Naegleria Fouleri
viii) Ringworm / Candidiasis

3. a) Heart attack
b) High BP
c) Digital BP Apparatus
d) Oxygen level
e) Body Temperature
f) Internal body condition
g) Internal organs
h) Internal organs

4. A - General medicine
B - Heart
C - Children
D - Pregnent women
E - Related to Urinary infections
F - Kidneys
G - Nervous system
H - Ear, nose, throat

5. Yes, use antibiotics only as prescribed by doctor only, otherwise it causes following side effects.
- Problems in the digestive system such as diarrhoea, vomitting and stomach pain
- Allergy
- Damage to the liver and kidneys
- Destruction of beneficial bacteria in the intestine
- Resistance against antibiotics (Antibiotic resistance)

6. a) Malaria, Plasmodium (Protozoa)
b) Candidiasis, Fungus
c) AIDS, HIV
d) Leptospirosis (Rat Fever), Leptospira

7.
- A rigid cell wall, epidermis and waxy coating on leaves and stems prevent the entry of pathogens.
- Antimicrobial substances either kill pathogens or inhibit their growth.
- Enzymes break down the cell walls of pathogens.
- The cells in the infected area undergo autolysis to prevent the spread of the disease.
- When one part of the plant is attacked, it sends warning signals from cell to cell about the disease.
- If pathogens cross the cell wall and enter, a polysaccharide called callose prevents their spread.

8. a) The process of transffering blood or its components from one person to an other is called Blood transfusion.

b) Individuals between the age of 18 & 60 can donate blood.
A minimum weight of 50kg is required

c) Blood groups are determined by the presence of A antigen and B antigen present on the surface of RBCs.

d) Antibodies a & b
Rh factor - Nil

9. a) Bud rot of coconut, Fungus

b) Mosaic disease in tapioca, Virus

c) Blight disease of Paddy, Bacteria

d) Bunchy top or banana, virus

10. a) Neutrophil, Eosinophil, Monocyte, Basophil

b) Phagocytosis, Inflammatory responses, Clotting of blood, Fever, healing of wounds.

Chapter 6

BIOLOGY AND TECHNOLOGY

1 Mark Questions

1. Correct the statements in the underlined words, if mistakes are there.
   a) The technology of using living things or their parts, to provide products and services that are useful to humans is called Genetic Engineering.
   b) Biotechnology is the technology for creating desired traits in organisms by combining or deleting genes.

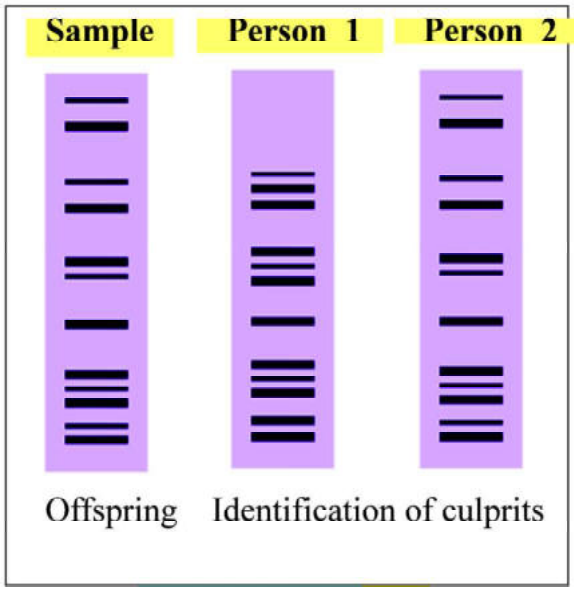
2. Choose the wrong statements related to Human Genome Project
   a) The exact sequence of 500 crores base pairs in the human genome
   b) Gene mapping technology identifies the location of genes
   c) 98% of genes are junk genes
   d) Human genome project was started in 2000 and was completed in 2004

3. Analyse the table and choose the correct answer

	Characteristics	Microorganisms
a)	Abililty to decompose wastes	i) Rhizobium
b)	Ability to produce insulin	ii) Baker's yeast
c)	High nitrogen fixing ability	iii) Psuedomonas putida
d)	Fementation ability	iv) E.coli

1) a. iii, b. iv, c.i, d.ii
2) a.ii, b.iv, c.i, d.iii
3) a. ii, b. i, c.iv, d.ii
4) a.iii, b. iv, c.ii, d.i

4. Analyse the illustration and answer the questions



a) Identify the technology in the illustration?

b) Find out the culprit from the illustration?

5. Who discovered DNA Fingerprinting technology?

a) Robert Hook
b) Sir Alex Jeffrey
c) James Watson
d) Francis Crick

6. Write the full form of HMP?

a) Human Microbe Project
b) Human Microorganisms Project
c) Human Microbiology Project
d) Human Microbiome Project

7. Complete the table

Characteristics	Examples
Pest resistance	B.T. Cotton
Weedicide resistance	Soyabean
Nutrient enriched	a) _____
Virus resistance	b) _____

8. Choose the correct statements related to Recombinant DNA technology

a) Recombinant DNA technology combines DNA segments from two or more organisms

b) Restriction endonuclease enzyme is used for joining DNA segments

c) Plasmid and viruses are used as gene vectors

d) Ligase enzyme is used to cuts a specific gene

i) a, b and c are correct ii) a and c are correct

ii) b and c are correct iv) a, b, c and d are correct

9. Find out the correct pair from the following options

a)	Wildlife tracking sensors	-	A technology for collecting biodiversity in formations
b)	Cryopreservation	-	Preserving living cells, tissues and organs at very low temperature
c)	Bioremediation	-	Remove pollutants from environment
d)	Global Biodiversity Data Sensors		Wild life conservation

i) a, d ii) b, c iii) b, d iv) a, b, c, d

10. Which of the following statements about recombinant DNA technology is correct?

i. Restriction enzymes are used to cut the DNA of an organism.

ii. Ligase enzymes join the cut DNA fragments together.

iii. Vectors are usually bacterial plasmids.

iv. When recombinant DNA enters a host cell, it does not replicate.

a) i and ii only b) i, ii, iii only

c) i, iii, iv only d) i, ii, iii, iv all

11. Name the new scientific discipline of science that analyses biological data by combining the latest technological disciplines such as computer science, mathematics and statistics.

a) Date Science b) Bioinformatics

c) Biotechnology d) Genetic Engineering

12. Identify A, B, and C and choose the correct option

A - A technology that joins together DNA segments from two or more different organisms

B - A technology used to locate genes in DNA

C - A technology used to identify individuals by analysing the arrangement of nucleotides in DNA

i) A - Recombinant DNA Technology, B - Gene Mapping, C - DNA Finger printing

ii) A - Human Genome Project, B - Gene Therapy, C - CRISPR Technology

iii) A - Recombinant DNA Technology, B - DNA Fingerprinting, C - CRISPR Technology

iv) A - Gene Therapy, B - Gene Mapping, C - Human Genome Project

13. Choose the modern technology used to edit DNA with great precision

a) DNA Technology b) CRISPR

c) Human Genome Project d) DNA Fingerprint technology

14. The steps involved in CRISPR Technology is given below. Arrange them properly.

[a) cutting b) Editing c) Identification of target]

a) a, b, c b) b, c, a

c) c, a, b d) b, a, c

2 Mark Questions

1. What are the differences between Biotechnology and Genetic Engineering?

2. Plants such as Bt Cotton, Soyabean, Golden rice and RSV - resistant papaya are genetically modified varieties.

   What are the specific characteristics obtained by these plants through genetic modification?

3. The application potential of Artificial Intelligence (AI) in the field of biology is very high. Write four areas in biology where AI is used to validate this statement.

4. Numerous biological techniques have been developed for the conservation of eco system and biodiversity. Write any four such technologies used.

5. Suggest four sub-topics for organizing a seminar on the topic 'Biologica Technologies : Potential and Challenges'.

3 Mark Questions

1. Describe and give an example of the tools used in biotechnology of forming recombinant DNA by joining DNA fragments from two or more different organisms.
   a) Genetic Scissors
   b) Genetic glue
   c) Vectors

2. Organisms whose genetic structure has been altered through genetic technology are called Genetically Modified Organisms (GMO). What are the specific characteristics introduced through genetic modification in Plants, animals, micro organisms?

3. Sickle Cell Anaemia is a heriditary disease. The World Health Organisation has approved Gene therapy for the complete cure of this disease. Explain different steps of stem cell therapy.

4. The Human Mircobiome Project was a project launched by the National Institute of Health in the USA in 2007.
   a) What is Human Microbiome?
   b) According to human microbiome project which are the organisms live in the human body?
   c) What services does the microbiome provide?

4 Mark Questions

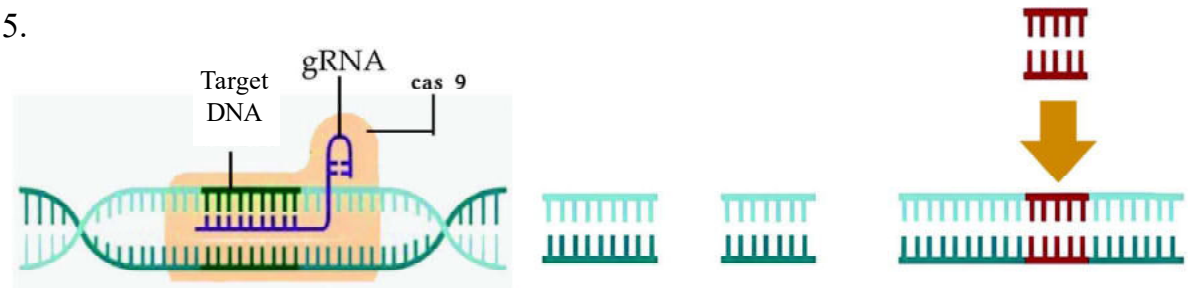
1. Artificial Intelligence (AI) plays a major role in the field of biological studies. Describe how it works in the following areas. Write your findings.
   a) Disease diagnosis and treatment
   b) Personalized medicine
   c) Genome sequencing
   d) Bio Informatics

2. Numerous biological techniques have been developed for the conversation of eco systems and bio diversity. What are the benefits of the following techniques.
   a) Wild life tracking sensors
   b) Global Biodiversity Data sensors
   c) Cryo preservation
   d) Bio remediation

3. Just as fingerprints are used to identify individuals, DNA Fingerprinting can also be used today.
   a) Prepare a definition for DNA Fingerprinting.
   b) How can individuals be identified through fingerprinting?
   c) What are the uses of DNA Fingerprinting?

4. The Human Genome Project was a large global scientific endeavor that began in 1990 and was completed in 2003 with the aim of fully understanding the information in the human genome.
   a) What are the facts discovered through the Human Genome Project?
   b) What is Genome and Gene Mapping?

5.



   a) What does the illustration indicate? Prepare a definition.
   b) What are gRNA and Cas 9 enzyme?
   c) What is Gene Editing?

Chapter 6

BIOLOGY AND TECHNOLOGY

ANSWERS

1 Mark

1. a) Biotechnology
   b) Genetic Engineering
2. a) The exact sequence of 500 crore bases pairs in the human genome
   d) Human genome project was started in 2000 and was completed in 2004
3. 1) a. iii, b. iv, c.i, d.ii
4. a) DNA Finger printing
   b) Person 2
5. a) Sir Alex Jeffrey
6. d) Human Microbiome Project
7. a) Golden rice
   b) RSV resistence papaya
8. ii) a and c are correct
9. ii) b) Cryopreservation - Preserving living cells, tissues and organs at very low temperature
   c) Bioremediation - Remove pollutants from environment
10. b) i, ii, iii only
11. b) Bioinformatics
12. i) A - Recombinant DNA Technology, B - Gene Mapping, C - DNA Finger printing
13. b) CRISPR
14. iii) c, a, b

2 Mark

1. Biotechnology is the technology that uses living organism or parts of their bodies to produce products and services beneficial to humans. The growth of Biotechnology accelerated with the discovery of the structure of DNA.

   Genetic Engineering is the technology used to develop desirable traits in organisms by adding or removing genes.

2. | | | |
   |---|---|---|
   | Bt Cotton | : | Pest resistance |
   | Soyabean | : | Weedicide resistance |
   | Golden rice | : | Nutrient rich |
   | RSV resistant Papaya | : | Virus resistance |

3. - Disease diagnosis and treatment
   - Peronalized medicine
   - Genome sequencing
   - Improving Agriculture
   - Pollution prevention
   - Bio informatics

4. Wild life tracking sensors, Cryo preservation
   Global Biodiversity Data Sensors, Bio remediation

5. - Biotechnology
   - Artificial intelligence
   - Opportunities
   - Challenges

3 Mark

1. a) Genetic Scissors : The name given to the enzymes that cut out specific genes.
      Eg:- Restriction endonucleases
   b) Genetic Glue : Enzyme used to join two genes.
      Eg:- Ligase
   c) Vectors : Entities that carry gene are called Vectors.
      Eg:- Plasmids in bacteria

2. Characteristics introduced in plants.
   - Pest resistance
   - Herbicide resistance
   - Nutrient - rich
   - Virus resistance

   Characteristics introduced in animals
   - Faster growth
   - Disease resistance
   - High productivity
   - Flurorescence

   Characteristics introduced in Micro organism
   - Ability to degrade waste
   - Insulin production capability
   - Fermentation capability
   - Improved Nitrogen Fixation capability

3. Step 1 : Collecting stem cells that form blood cells
   Step 2 : Using a virus as a vector, the functional gene is incorporated in to the stem cells.
   Step 3 : The stem cells with the integrated gene are introduced back in to the patient.

4. a) Human Microbiome : The collection of micro organisms and their gene that live in an interact with the human body.
   b) Organisms : It is the community of micro organisms such as bacteria, archea, fungi and viruses. There are more micro organisms inside the human body.
   c) Services : The Microbiome provides many services such as immunity, digestion, production of vitamins and absorption.

4 Mark

1. a) Disease diagnosis and treatment : Used to finding new medicines, predicting the efficiency of chemical substances and suggesting the most appropriate treat ment methods.
   b) Personalized medicine : Analysing the genetic structure and health information of each individual to recommend the most suitable treatment methods for them.

c) Genome sequencing : Utilized for gene editing technologies to treat genetic disorders.

d) Bio information analysing large amounts of genetic information and suggesting simplification strategies.

2. a) Wildlife Tracking Sensors: Help wildlife protectors and researchers monitor the movement, behavior, and location of wild animals to plan conservation strategies

b) Global Biodiversity Data Sensors: Technology that collects information about biodiversity using methods like satellite remote sensing, acoustic analysis, and camera traps.

c) Cryopreservation: The process of preserving living cells, tissues, and organs at very low temperatures (-196^0 C). It is used for long-term storage in fields such as medicine, agriculture, and research.

d) Bioremediation: Uses microorganisms and plants to remove pollutants from contaminated environments such as soil, water, and air. This converts the pollutants into harmless compounds.

3. a) DNA Fingerprinting: A technology used to identify individuals by analyzing the arrangement of nucleotides in the DNA.

b) Identification: Just as fingerprints are different, the arrangement of nucleotides in DNA is also different. This feature helps in identifying individuals.

c)
- Helps to identify criminals.
- Helps to identify genetic disorders and study hereditary diseases.
- Helps to determine the biological relationship between parents and children.
- Helps to identify fossils and understand lineages.
- Helps to identify people involved in accidents or disasters.
- Helps to track endangered species.

4. a) Facts Discovered: The exact sequence of 3 billion base pairs in the human genome.

Humans have between 20,000 and 25,000 genes.

Only about 1-2% of the human genome consists of functional genes. The remaining large portion is known as junk DNA.

b) Genome: The totality of an organism's genetic material.
Gene Mapping: The technology used to find the location of genes in the DNA.

5. a) Illustration: Indicates CRISPR Technology. Definition: It is a technology used to edit DNA with high precision.

b) Components : gRNA (Guide RNA): Locates the exact part of the DNA to be cut.
Cas 9: This is an enzyme; it is used to cut the DNA.

c) Gene Editing : The process of editing the arrangement of nitrogen bases in DNA in a way required to change an organism's characteristics. Through this, an undesirable trait can be removed or a new trait can be added.

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